

Racial Disparities in the U.S. Mortgage Market: Evidence from Data Privacy Legislation *

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Abstract

We study the unintended effects of data privacy legislation (DPL) on racial disparities in U.S. mortgage lending. Using the staggered enactment of state-level DPLs, we find that DPL reduces the mortgage denial gap between minority and non-minority borrowers by 3% and the interest rate gap by 8 bps. These effects are driven by increased automation and standardization in mortgage underwriting and FinTech lending, particularly in areas with high minority population, low bank branch density and high racial animus. Consistent with improvements in credit allocation not due to borrower composition, ex-post delinquency gaps decline while applicants' ex-ante credit risk remains unchanged.

JEL Classification: G21, G28, G18; J15; J71; O33

Keywords: Data Privacy Legislation; FinTech; Mortgage lending; Race; Financial Inclusion

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Abstract

We study the unintended effects of data privacy legislation (DPL) on racial disparities in U.S. mortgage lending. Using the staggered enactment of state-level DPLs, we find that DPL reduces the mortgage denial gap between minority and non-minority borrowers by 3% and the interest rate gap by 8 bps. These effects are driven by increased automation and standardization in mortgage underwriting and FinTech lending, particularly in areas with high minority population, low bank branch density and high racial animus. Consistent with improvements in credit allocation not due to borrower composition, ex-post delinquency gaps decline while applicants' ex-ante credit risk remains unchanged.

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1. Introduction

The recent wave of data privacy legislation (such as the EU’s GDPR (General Data Protection Regulation) in 2016 and the California Consumer Privacy Act (California Consumer Privacy Act) in 2018) has reshaped how personal data is collected and used across industries. In the U.S. mortgage sector, where personal data determines loan approval and interest rate decisions, these laws may have far-reaching and unintended consequences. Several U.S. states have introduced similar privacy laws in recent years (e.g., Virginia, Indiana and Texas), coinciding with a period of rapid automation in mortgage underwriting. Although prior studies suggest a decline in racial gaps due to legislations¹ and automation of underwriting processes, disparities in mortgage approval and interest rates between different groups of borrowers persist even after accounting for borrowers’ economic characteristics (e.g., Munnell et al., 1996; Bartlett et al, 2022; Gerardi et al., 2023; Das et al., 2023).² Moreover, there are enduring concerns that algorithmic decision making may entrench bias rather than eliminate it (CFPB, 2022). We show that, contrary to these concerns, data privacy legislation (DPL) can reduce racial gaps in mortgage denial and pricing, primarily by accelerating automation and standardization and expanding FinTech lending in underserved areas. We thus shed new light on how data privacy regulations affect equity in credit markets, particularly for racial and ethnic minority borrowers.

In this paper, we examine the impact of DPL on mortgage lending outcomes for minority and non-minority borrowers. Our findings suggest that state-level privacy laws can lead to a reduction in disparities between these groups by narrowing the denial and interest rate gap between minority

¹ The U.S. enacted Title VIII of the Civil Rights Act of 1968 (Fair Housing Act), which "prohibits discrimination in the sale, rental, and financing of dwellings [...] based on race, color, religion, sex, familial status, national origin, and disability."

² Bartlett et al. (2022) show that Latinx/Black borrowers, given similar risks, pay substantially higher rates for GSE-securitized and FHA-insured loans, particularly in communities with high minority populations. Similarly, Bayer et al. (2018) found that after controlling for credit scores and other risk factors, African American and Hispanic borrowers were 103% and 78% more likely, respectively, to receive high-cost mortgage loans. However, differences can be driven by factors other than race, such as borrower liquidity or preferences (cf. Bhutta and Hizmo, 2021; Howell et al., 2024; Bhutta et al., 2024; Gao et al., 2024).

and non-minority borrowers, affecting both the extensive margin (loan approvals) and the intensive margin (loan pricing). We argue that these effects are driven by a shift toward more automated and standardized underwriting processes, particularly in areas with high minority population, low bank branch density and high racial animus, while FinTech lenders further reduce disparities due to their broader geographic reach. Although DPL does not affect the ex-ante risk profile of applicants, the reduced ex-post delinquency gap among originated borrowers indicates that the shift toward more automated and standardized underwriting improves lenders' credit risk assessments.

Several U.S. states have enacted DPL since 2018, one notable example being the CCPA, which was signed into law in June 2018. DPL generally emphasizes three key aspects. First, DPL mandates transparency regarding how businesses collect, use, and share personal information, informing consumers about the types of data collected, the purpose of collection, and whether the data will be sold or shared. Second, consumers gain more control over their personal information through rights such as the right to know, the right to delete, the right to opt-out, and the right to non-discrimination when exercising these rights. Third, DPL imposes strict regulations on businesses to ensure the security and integrity of data post-collection, requiring businesses to protect customer data and prevent data breaches/security incidents. These legal frameworks have contributed to increased public awareness and usage of data right, reflecting a broader societal shift toward prioritizing data privacy.³

Using HMDA mortgage application data from 2018 to 2023, our main sample focusses on counties in U.S. states that have enacted DPL as the treatment group, and neighboring uncovered counties (that do not belong to states that have enacted DPL but border on counties that do) as the

³ See details in: <https://www.oag.ca.gov/privacy/ccpa>. According to McKinsey (2020), consumers have become more attentive to the types of data they share and with whom they share it. Particularly financial services and healthcare rank among the most trusted sectors for data sharing—indicating both heightened expectations and exposure to privacy standards.

control group, including information on mortgage market application, rejection rates, and racial origin data. Using a Difference-in-Differences approach, we examine disparities between minority and non-minority mortgage applicants across various states, before and after the enactment of DPL. Our main findings show that the denial gap between minority and non-minority applicants decreases by 3% and the interest rate gap narrows by 8 bps. We find that the results are driven by an increase in assessments of mortgage applications using an automated underwriting system (AUS) and using standardized credit models. Furthermore, effects are more pronounced for banks with high IT investment and FinTech lenders and in areas with a high level of racial bias and low bank branch density. FinTech lenders experience an overall increase in mortgage applications after enactment of DPL and widen their mortgage offerings (measured by the number of accepted mortgage applications and the loan amount) to both minority and non-minority lenders. Although applicants' ex-ante credit risk remains unchanged, the racial gap in ex-post delinquency among funded borrowers becomes smaller after DPL, indicating that DPL improves the accuracy of credit screening rather than altering applicant risk.

We apply the methodology by Bertrand and Mullainathan (2003) to examine the dynamic and stacked effects of DPL on racial disparities in mortgage lending to address concerns about reverse causality and to validate the parallel trends assumption. While we do not find any changes in credit risk of mortgage applicants before and after DPL, we further apply propensity score matching (PSM) to match mortgage applications with similar credit characteristics in treatment and control counties, to rule out any confounding effects of credit risk. Additionally, we conduct subsample analyses based on individual states and different privacy laws, finding more pronounced effects for legislations highlighting data transparency. To rule out any biases, we assess the lagged effects of DPL, omit samples from states where mortgages were signed in the same year the privacy laws

were enacted and exclude samples from the COVID-19 period (2020-2021) and samples from the top ten mortgage lending institutions. All models consistently confirm a reduction in denial and interest rate gaps between minority and non-minority groups after DPL.

We highlight automated underwriting and process standardization as the main mechanisms for the asymmetric implications of DPL on minorities' and non-minorities' access to and cost of credit. The emphasis on data transparency and control of U.S. DPL compels lenders to adapt their underwriting processes to comply with regulatory requirements, shifting from relationship-based assessments to standardized assessments and automated processes, thereby reducing the role of human discretion and discriminatory practices. We rule out shifts in borrower characteristics or credit demand as changes in borrower credit risk or loan applications after DPL are insignificant.

We further illustrate heterogeneous effects with respect to lender types and geographical region. Effects are more pronounced for lenders with high IT investment and FinTech lenders, which also tend to rely more on automated screening processes and data-driven assessment rather than relationship lending, with FinTech lenders exhibiting an additional 2.7 percentage point reduction in minority denial gaps and a further 4.4 bps decline in interest-rate gaps relative to traditional lenders. While we do not observe an overall credit expansion for all lenders, we report an increasing number of loan applications and approved mortgages by FinTech lenders after DPL, as their standardized and faster lending processes (Fuster et al., 2019) and independence from traditional branch networks can facilitate the mortgage application process after DPL. We further report wider mortgage offerings by FinTechs in minority-dense or underbanked areas, thereby enhancing access to finance in underserved communities.

Our contribution is threefold. First, we contribute to the emerging literature on data security and DPL and its effects on the banking industry, particularly on mortgage lending, by providing

novel empirical insights into the impact of DPL on racial disparities. Existing work has focused on the effects of the CCPA on mortgage lender and borrower costs (Gupta et al., 2023), on FinTech lending (Doerr et al., 2023), and the impact of U.S. privacy laws on small banks' IT investment (Huang et al., 2025). Gupta et al. (2023) show an increase in data processing expenses (especially for soft information) after the CCPA has come into effect, while Huang et al. (2025) demonstrate pre-emptive IT investments of small banks after DPL proposals. Doerr et al. (2023) further show an increase in data sharing due to decreased customer concerns about data security and data sharing. In our study, we focus on the impact of a wider range of data privacy laws on racial disparities in mortgage lending, driven by both automated and hard information-driven underwriting processes as well as wider FinTech lending in underserved areas.

Second, we contribute to the discourse on racial disparities in the mortgage market by offering new evidence on the unintended but asymmetric effects of data privacy laws (that do not target racial disparity), extending insights of existing studies on racial disparities (see Munnell et al. (1996), Bayer et al. (2018), Ambrose et al. (2021) and Bartlett et al. (2022)). Recent studies highlight the role of algorithmic underwriting and process automation, which have the potential to decrease racial discrimination and bias for both small business lending (Howell et al., 2024) and mortgage markets (Gao et al., 2024; Bhutta et al., 2024). While Gao et al (2024) study the introduction of automated underwriting systems for riskier (high Debt-to-Income) loans insured by the Federal Housing Administration (FHA), Bhutta et al. (2024) study the effects of automated underwriting on a wider set of mortgages. In contrast, we focus on a more homogenous sample of conventional loans guaranteed by government-sponsored enterprises (GSE), Freddie Mac and Fannie Mae⁴, with a stable distribution of Debt-to-Income (DTI) characteristics before and after

⁴ As noted by Gupta et al. (2024), DPL such as CCPA induces a shift to originate-to-distribute model, from the traditional banking model.

DPL. While we do not report significant effects on delinquency as Gao et al. (2024), we illustrate asymmetric effects of DPL on minority and non-minority applicants, illustrating differential effects to a pure expansion of automated underwriting. While we rule out confounding effects due to credit risk, we conclude that the reduction in denial and interest rate gaps between minority and non-minority borrowers after DPL is further driven by both automation and the increasing standardization of lending processes.

Finally, our study contributes to the wider exploration of FinTech in mortgage lending and underwriting⁵, as evidenced by Buchak et al. (2018), Fuster et al. (2019), Parlour et al. (2022), and Das et al. (2023), among others. FinTech lenders are playing an increasingly important role in mortgage lending, as they use online applications and automated processes to offer faster loan evaluations (Fuster et al., 2019) and attract more minority applicants (Chernenko et al., 2022). We find that DPL has important effects on FinTech and traditional lending for minority and nonminority borrowers. Applicants' observable credit risk does not change with DPL, but the narrowing delinquency gap among actual borrowers implies that DPL leads to decreasing bias between minority and non-minority lenders and more consistent assessment of credit risks during the underwriting process.

Our results hold important policy implications, as DPL does not specifically target racial disparity but has unintended beneficial effects for minority borrowers in the form of reduced denial and interest rate gaps, as well as reduced ex-post delinquency disparities despite unchanged ex-ante applicant risk, and heterogeneous effects on traditional banks and FinTech lenders. By expanding credit access to borrowers with thin credit files and in underserved communities, DPL and its implications for mortgage lending processes contributes to efforts aimed at reducing

⁵ See Berg et al. (2022) for an excellent overview of the FinTech lending literature.

structural barriers in the mortgage market, contributing to a wider discussion on the social implications of digitalization and automation of financial services and wider financial inclusion.

The rest of the paper is organized as follows: Section 2 provides further institutional details and develops our main hypotheses. Section 3 provides details of our data sample and outlines the empirical methodology. Section 4, 5 and 6 present the results, mechanism tests, and heterogeneity tests. Section 7 finally concludes.

2. Institutional Background and Related Literature

2.1 Institutional Background

Since 2018, various U.S. states have introduced and enacted consumer privacy regulations that require entities to ensure the security and integrity of customer data as well as strengthen consumers' control over their data, increasing transparency and accountability in data handling. California led the way with the California Consumer Privacy Act (CCPA), which was passed in early 2018 and signed into law on June 12 of the same year. The CCPA is specifically designed to protect the privacy rights and consumer rights of California residents, mandating data protection and privacy obligations for businesses that collect, process, or sell the personal information of California residents. The CCPA imposes strict regulations on these companies to ensure the security and integrity of collected data. It further introduces new rights for California residents, including (1) the ability to access and delete personal information, (2) the option to opt out of selling personal information to third parties, and (3) the right to non-discrimination when exercising these rights. These requirements come with significant penalties in case of non-compliant behavior. Depending on the nature of the violation, penalties under the CCPA can

include civil fines of up to \$7,500 per violation, along with additional fines ranging from \$100 to \$750 per consumer in the event of a data breach.

[Insert Figure 1 about here]

As shown in Figure 1, 12 states have signed DPL by the end of 2023 to address recent data breaches and privacy protection issues, requiring companies to develop and implement mandatory protective tools to safeguard private information (Huang et al., 2025). Figure 1a shows the states that have enacted DPL (treatment group): darker blue indicates earlier enactment, while lighter blue represents more recent enactments. Adjacent (control group) states are indicated in grey. For our analysis, we focus on the counties on the borders of states with DPL (treatment) and without DPL (control), as depicted in Figure 1b. While specifics of each state's DPL differ, all states include the basic rights to access, delete, opt-out, and non-discrimination (same as CCPA). Details of the enactment dates and provisions of each state's DPL are provided in Appendix Table A1.

Following the enactment of these laws, consumer awareness and concern regarding personal data protection increased substantially. As shown in Figure OA2, public interest in DPL surged, with Google search trends for terms such as "California Consumer Privacy Act" and "CCPA" spiking noticeably around the announcement and implementation periods of DPL. In addition, DPLs have been actively enforced, with non-compliance resulting in substantial financial penalties and reputational risk for firms.⁶ In compliance with DPL, financial institutions have thus become more cautious in handling customer data. As shown in Figure OA3, consumer complaints related to the misuse of personal information declined following the implementation of DPLs, indicating that these laws have had a positive and measurable impact on data governance and accountability.

⁶ For example, the California Attorney General's Office fined Wells Fargo \$7.6 million for recording customer phone calls without proper disclosure, in violation of state privacy laws. Similarly, Citibank was fined \$420,000 after a technical vulnerability led to the exposure of personal data belonging to over 80,000 account holders. As part of the settlement, the bank was required to enhance its cybersecurity protocols and provide credit monitoring services to affected consumers. See here for more details: <https://oag.ca.gov/privacy/privacy-enforcement-actions>

2.2 Racial Disparities in Mortgage Lending and Implications of DPL

Significant racial disparities persist in mortgage lending, as minority applicants experience higher annual interest rate spreads and are more frequently denied loans than their white counterparts (Black et al., 2003). Bartlett et al. (2022) show that risk-equivalent Latinx/African-American borrowers, given similar risks as white borrowers, pay substantially higher rates for GSE-securitized and FHA-insured loans, particularly in communities with high minority populations. Similarly, Bayer et al. (2018) found that after controlling for credit scores and other risk factors, African American and Hispanic borrowers were 103% and 78% more likely, respectively, to receive high-cost mortgage loans.⁷ Ambrose et al. (2021) found that mortgage brokers charged premiums to borrowers outside their ethnicity, with Hispanic brokers showing favoritism toward Hispanic borrowers. For Paycheck Protection Program (PPP) loans, Chernenko et al. (2024) and Howell et al. (2024) observed greater racial disparities in financing access in counties with higher racial animus, suggesting loan decisions may be influenced more by geographical location or lender biases than by individual default risk.

However, Bhutta et al. (2024) show that observed and unobserved risk factors, rather than discrimination, can account for significant disparities in denial, but find that lenders provide lower-quality service to minority borrowers. Similarly, mortgage institutions have been accused of "redlining" minority communities rather than individuals, leading to differential treatment of loan applications from certain neighborhoods and offering fewer services and branches in these areas (Holmes and Horvitz, 1994; Tootell, 1996).

⁷ Distinguishing whether these racial disparities stem from statistical discrimination, where lenders observe attributes unknown to researchers - or from direct taste-based discrimination, bias, and stereotypes is challenging. If statistical discrimination were the primary cause of this discrepancy, loan costs would be unrelated to race. However, even after controlling for various credit characteristics, minority borrowers still incur significantly higher costs compared to white borrowers (Bayer et al., 2018; Gerardi et al., 2023). In contrast, if stereotypes and bias drive these differences, loan costs would correlate directly with race.

As DPL introduced strict requirements for data processing and consumer data rights, the transparency requirements of privacy statutes can compel financial institutions to standardize their underwriting processes. Banks traditionally held an information monopoly on their clients and were able to infer soft information, for example from personal interactions. These interactions provided valuable insights into borrower's credit worthiness, thereby enhancing default prediction accuracy (Berg et al., 2020; Iyer et al., 2016) and reducing information asymmetry (Berger et al., 2005; Agarwal et al., 2011). However, reliance on soft information can also lead to biases, as credit officers may respond to perceived race or ethnicity based on an applicant's name or location (Begley and Purnanandam, 2021) and could potentially disadvantage minority candidates in evaluations and mortgage costs (Ambrose et al., 2021).⁸

While DPL's original purpose focusses on data transparency and protection, we conjecture that DPL has asymmetric (unintended) effects for minority borrowers compared to non-minority borrowers. Due to the increased costs of soft information compliance (Gupta et al., 2025) and relationship lending via bank branches and face-to-face interactions, DPL may lead to increasing reliance of lenders on automated and standardized lending processes. The shift towards automated and standardized underwriting has the potential to reduce the impact of human credit officer biases, particularly benefiting minority borrowers (Bartlett et al., 2022; Howell et al., 2024).⁹ Based on these considerations, we propose the following hypothesis:

H: Data privacy legislation significantly reduces racial disparities in mortgage lending at both the extensive margin (access to loans) and intensive margin (loan costs).

⁸ Gao et al. (2024) show that a credit expansion using algorithmic underwriting by the Federal Housing Administration (FHA) in 2016 had heterogeneous effects on different types of borrowers and overall welfare improvements for disadvantaged borrowers, but limited effects on default risk.

⁹ For instance, standard credit scoring models such as the FICO Risk Score and Vantage Score are utilized and developed by companies specializing in proprietary algorithms. Lenders and investors rely on these scores as indicators of credit risk. For example, FICO scores range from 400 to 900, with 650 being the widely relied-upon minimum score for mortgage eligibility, see <https://www.fico.com/en/products/fico-score>.

2.3 DPL Implications for Automated and Standardized Mortgage Underwriting

As the recent wave of state-level DPL mandates substantial enhancements in the data protection capabilities of organizations, financial institutions are faced with increasing compliance costs in the form of increased IT investment, as well as legal, data processing, and telecommunication expenses.¹⁰ Furthermore, DPL has strengthened the requirements for the management and disclosure of personal information collection, processing, and sharing, ensuring transparency and compliance in data handling. These regulations compel financial institutions to provide information in user-friendly formats, particularly requiring banks to invest in hardware and IT infrastructure to meet these standards (Huang et al., 2025). Due to the increasing compliance cost of soft information (cf. Gupta et al., 2023), we further argue that DPL will induce a further shift towards automated mortgage underwriting. The impact of DPL on minority mortgage rates and rejection rates should thus depend on the institution's reliance on automation and process standardization.

Reliance on automation and standardization might not necessarily lead to more objective lending processes, as the automation of loan decisions can extend human biases to algorithms.¹¹ While aspects of algorithmic bias and fairness are beyond the scope of our paper, there are concerns that historical decisions encumbered by human bias and used as training data for algorithms might further exacerbate racial stereotypes through automated mortgage lending processes. However, empirical evidence is sparse and suggests a rather positive effect on minority borrowers. Das et al. (2023) provide a detailed overview of the literature on algorithmic fairness

¹⁰ Gupta et al. (2023) quantify the compliance costs associated with the introduction of the CCPA, highlighting the greater costs for banks that rely more on soft information and a shift in market share from banks to nonbanks. These increased compliance costs can be passed through to bank customers through increased mortgage rates and reduced loan availability due to bank retrenchment.

¹¹ Morse and Pence (2021) provide a general overview of the ambiguous effects of innovation and data privacy laws on (both taste-based and statistical) discrimination in household finance.

and provide evidence of the differential role of human and algorithmic underwriting on female/male and minority/nonminority borrowers, although finding mostly fair treatment across groups. Evidence by Howell et al. (2024) and Gao et al. (2024) suggests that automation and algorithmic underwriting lead to increases in PPP lending to Black borrowers and in mortgage lending to White and Hispanic borrowers, respectively. Furthermore, Jansen et al. (2024a) do not find a significant bias of algorithmic underwriting in funding rates for racial minorities, but rather a modest positive bias towards Hispanic workers.

2.4 The Role of FinTech Lending

Recent literature on DPL and open banking (the process of opening bank customer data and making it more accessible and portable at the request of customers) has shown that DPL and increased data portability favor FinTechs rather than traditional banks (Parlour et al., 2022; He et al, 2023; Doerr et al, 2023; Gupta et al., 2023). Thus, the shift in market share can affect minority and non-minority borrowers differently, as some FinTech lenders rely more heavily on automation than traditional lenders (DiMaggio and Yao, 2021; Howell et al., 2024). There are significant differences among lending institutions in terms of information processing capabilities and underwriting processes: Banks tend to use a mix of hard and soft information (such as judgments of loan officers) in making lending decisions, while FinTech lenders rely more on automated processes and hard information to determine mortgage rates (Fuster et al., 2019).

There is mixed evidence regarding the effects of FinTech lending on minorities' access to credit. Barlett et al. (2021) find that effects depend widely on the type of mortgage, particularly for FHA loans, minority borrowers experience a reduction in rate premia for FinTech loans, and thus a potential reduction in discrimination. There are further concerns about whether the widening

of credit access to minority borrowers means including low-credit score borrowers and increasing credit risk, as shown for the consumer loan market by DiMaggio and Yao (2021). However, Fuster et al. (2019) find no increase in riskier mortgage loans issued by FinTechs. Rather than widening borrowing to low-quality lenders, Buchak et al. (2018) show that nonbank lenders focus on more creditworthy borrowers, although they seem to use different information to set interest rates relative to other lenders, pointing towards better pre-screening abilities of nonbanks. Several studies further show the potential of Machine Learning algorithms and Robo-advising to improve lending efficiency (Tantri, 2021) and reduce discrimination (D'Acunto et al., 2021).

The greater role of FinTech lenders can also have beneficial effects for underserved communities, as FinTechs can operate through online platforms, providing services without geographic constraints, while traditional banks rely on branch networks which are often sparse in minority communities (Howell et al., 2024). Due to their entirely online and more efficient loan processing, FinTechs can adapt quickly to market and policy shifts (Buchak et al., 2018; Fuster et al., 2019) and potentially serve minority-dominated areas underserved by traditional banks.

We thus conjecture that DPL will further increase competitive pressure on traditional lenders by FinTechs. As FinTech lenders rely more on automated underwriting and standardized processes and are less constrained by geographical restrictions such as bank branch availability, they are able to provide wider offers and more services to traditionally underserved communities at potentially lower interest rates than traditional lenders. DPL can thus lead to expanded credit options for minority borrowers who might otherwise face redlining in traditional lending. DPL grants consumers more rights to access and share their data, improving their loan choices and opt-out options. As a result, borrowers are no longer limited to traditional banks and can more easily apply for loans from nonbank lenders and FinTechs.

However, some theoretical studies caution against the unintended consequences of increased data portability and data sharing (through initiatives such as open banking) on borrowing rates (Parlour et al., 2022) and bank performance (Lin et al., 2024). Parlour et al. (2022) caution against the disruption of information spillovers within traditional banks due to the shift of customer data to FinTechs and subsequent increases in information asymmetry and interest rates. He et al. (2023) find that open banking can lead to over-empowerment of FinTechs, which can lead to a widening of the screening ability gap, as some customer information might only be available to FinTechs. Lin et al. (2025) find heterogeneous effects for banks that adopt bank APIs to share customer data, as smaller banks experience improvements in value and growth prospects, while direct benefits to large banks are limited.

3. Data and Methodology

3.1 DPL and Identification Strategy

Since 2018, various U.S. states have enacted DPL. We gather information on consumer DPL enacted in the U.S. between 2018 and 2023, provided by the International Association of Privacy Professionals (IAPP). In our empirical analysis, we divide mortgage applications into treatment and control groups to assess the impact of DPL on racial disparities in the mortgage market., as presented in Figure 1a. Applications in the states that enacted consumer DPL constitute our treatment group (in blue), whereas those submitted in adjacent states without such legislation form our control group (in grey).

To ensure a clearer identification and distinction between counties that are affected and counties that are unaffected by DPL, we further restrict our sample to include only counties on the comparative borders of states with DPL (treatment) and without DPL (control). This identification

strategy (as shown in Figure 1b) takes advantage of the similarity between applicants in neighboring counties over randomly selected households across counties, eliminating potential confounding variables by maintaining relatively constant economic environments between the treatment and control groups. Moreover, this method accounts for changes over time between counties and races, such as local economic conditions or credit market situations.

3.2 Mortgage Lending Data

To construct our sample, we integrate multiple data sources. We provide an overview of these datasets below. We collect comprehensive mortgage application data from 2018 to 2023 from the Home Mortgage Disclosure Act (HMDA) database, and loan origination and performance data from the government-sponsored enterprises (GSEs), including Fannie Mae Freddie Mac. HMDA includes detailed application information on borrowers and the properties mortgaged (down to the census tract, county, and state levels). This dataset covers 90% of mortgage originations in the U.S. and provides detailed information on loan rates, spreads (relative to the average prime offer rate for comparable mortgages), costs (expressed as a ratio of total settlement costs to the loan amount), purpose (for purchase or refinancing), loan amount, term (in months until maturity), loan-to-value ratio (LTV), potential property value, property location by census tract, qualifying loan status, lien status, and originator identity. Additionally, specific borrower information such as gender, race, and age is included.

We focus on conventional loans guaranteed by GSEs. We apply common data filtering procedures (Buchak et al., 2018; Gupta et al., 2024), including only loans with conforming loan limits (in line with GSE mortgage purchasing limits (Loutskina and Strahan, 2015)), first-lien loans

(excluding junior liens), and retaining loans aimed at home purchases, refinancing (including cash-out refinancing) and excluding reverse mortgages.

To assess the performance of loans in our dataset, we further match HMDA data with loan-level origination and performance data from Fannie Mae and Freddie Mac. The matching is based on property location (state and metropolitan statistical area), origination year, exact loan amount, exact interest rate, loan term, loan purpose, owner-occupancy status, and borrower characteristics such as the debt-to-income ratio and the combined loan-to-value ratio. Following Buchak and Jørring (2021), we ensure the highest match quality by excluding all duplicate loan observations and performing matching without replacement. A key advantage of the GSE datasets is that they provide direct measures of borrower risk, including two ex-ante indicators (FICO credit score and loan-to-value ratio) and an ex-post indicator based on mortgage delinquency. In addition to underwriting details, the Fannie Mae data provide monthly updates on loan performance, including delinquency status. Delinquency is defined as a loan being more than 60, 150, or 180 days past due within first one or two years of origination, consistent with prior literature.

We construct our main sample from counties in U.S. states that enacted DPL as the treatment group and uncovered counties that belong to states that NOT enacted DPL but are neighboring to covered counties as the control group (see Figure 1), including information on mortgage market application, rejection rates, and racial origin data. Our neighboring county sample comprises 7,222,834 mortgage application observations, which leverages the homogeneity between applicants from adjacent counties and families randomly selected across county lines, enhancing the reliability of the comparative analysis. Additionally, for robustness tests, we include an extra sample from states that are adjacent to states where DPL took effect, totaling 35,015,971 mortgage application observations.

Minority Borrowers

Following the research by Bartlett et al. (2022), we define borrowers of Latinx/Hispanic or Black descent as minorities, historically the most discriminated against groups in housing and financial sectors. Conversely, Asian Americans are excluded from this minority classification due to their generally higher wealth levels compared to other racial groups, with a median net worth significantly higher than that of White, Black, and Hispanic households.

FinTech Lenders

We further match the lender names and LEI IDs with RSSD IDs using HMDA data and follow the methodology of Buchak et al. (2018) to classify lending institutions into (1) traditional banks and (2) FinTech lenders. We identify FinTech lenders by checking whether the financial institution offers deposits and whether mortgage applicants can obtain pre-approval online, as well as whether loan operations are entirely online with no human intervention. FinTech lenders operate differently from traditional banks, often relying on alternative data sources for credit assessment and excelling in big data analysis (Buchak et al., 2018). To ensure robustness and minimize subjectivity in our classification, we also conduct cross-verification to overcome subjectivity in categorization. Specifically, we compare our FinTech lender classification with the list presented by Fuster et al. (2019) and confirm that our categorization remains consistent and reliable.

3.3 Methodology

To examine the impact of the DPL on racial disparities in the mortgage market, we combine data on mortgage market applications, rejection rates, and racial origins for our treatment group (neighboring covered counties belonging to states with enacted privacy protection laws) and

control group (uncovered counties belonging to states without such laws). We primarily estimate the following Diff-in-Diff model as shown in Equation (1):

$$y_{i,l,t,c} = \beta DPL_{ct} \times Minority_i + \theta DPL_{ct} + \mu Minority_i + Control_{i,t,c} + \varphi_{lt} + \omega_{lc} + \epsilon_{i,l,t,c} \quad (1)$$

where i , l , t , and c represent the loan application, lender, year, and county, respectively. The coefficients of interest pertain to the sequence of estimates associated with the interaction terms β . The independent dummy variable DPL_{ct} indicates whether the mortgage application pertains to a property located in a county (state) where DPL has been signed and has been submitted after the signing. $Minority_i$ is a dummy variable equal to one if the mortgage applicant belongs to a minority group (Black or Hispanic) and zero otherwise.

We focus on two dependent variables $y_{i,l,t,c}$: *Interest Rate* $_{i,l,t,c}$ - the loan interest rate in percent and *Denial* $_{i,l,t,c}$ - the denial situation which reflects whether the mortgage application is rejected by lenders or not. For our control variables, we include borrower characteristics, in particular *Credit Model* (whether a credit model was used to make the credit decision), *Debt-to-Income* (DTI) Ratio, *Loan-to-Value* (LTV) Ratio, *Loan-to-Income* (LTI) Ratio, *Loan_Term*, *Race Observed*, and *Age*, to control for possible determinants of default risk, which might influence the loan approval and loan pricing decision. Table OA1 delineates all control variables used in our analysis. We include lender $\times$ year, and lender $\times$ county fixed effects (φ_{lt} , ω_{lc}). Standard errors are clustered at the state level.

3.4 Summary Statistics

Table 1 presents descriptive statistics for the variables used in our analysis. We winsorize all continuous variables at the 1st and 99th percentiles to mitigate the impact of potential outliers. Our sample includes 7,222,834 mortgage application observations from counties with enacted DPL

and their adjacent counties without DPL (as depicted in Figure 1b) and 35,015,971 observations from states with enacted DPL and their adjacent states without DPL (as depicted in Figure 1a). The average value of the *DPL* variable is 0.187, indicating that 18.7% of mortgage applications observed have been submitted in counties within states that have enacted significant DPL. The average for minority applications is 0.178, suggesting that 17.8% of loan applications are submitted by minority individuals.

[Insert Table 1 about here]

In addition, we present visual evidence of the impact of the DPL on mortgage lending across different ethnic/racial groups. As shown in Figure 2, we compare the average interest rate spread (the difference between the mortgage rate and Annual Percentage Rate) and the loan application rejection rates for minority and non-minority groups over the three years before and after the enactment of the act in the treatment samples (which include applications submitted in states that enacted DPL during the sample period). The x-axis marks the years relative to the enactment of the DPL, with "Year 0" denoting the year the act was officially passed. Blue lines represent minority groups, while red lines represent non-minority groups. Evidence indicates that before DPL, interest and rejection rates were significantly higher for minority groups relative to their non-minority counterparts, with both rates exhibiting parallel trajectories. However, after the passing of state-level DPL, there has been a noticeable reduction in these disparities.

[Insert Figure 2 about here]

4. Empirical Results

4.1 Baseline and Dynamic Results

We analyze the impact of U.S. DPL on racial disparities in the mortgage market in terms of denial rates of loan applications and the offered loan interest rates. We compare applications submitted in treatment counties within states that have enacted DPL (covered counties) and control counties without DPL (uncovered counties) in Table 2. We report results based on the full sample in Panel A and based on the merged sample of accepted mortgage applications and GSE-securitized mortgages (including credit scores of GSEs as controls) in Panel B.

[Insert Table 2 about here]

In Panel A Columns (1)–(2), we assess whether DPL can reduce disparities in access to finance by a relative decrease in the rejection rates for minority loan applicants compared to non-minority applicants, focusing particularly on samples from states with and without enacted DPL. The β coefficient illustrates the differential impact on minority versus non-minority mortgage denial rates and interest rates before and after DPL enactment. Column (1) presents results from model (1) with fixed effects but without control variables, while model (2) includes borrower-specific controls, such as credit risk-related variables (*Debt-to-Income*, *Loan-to-Value*, *Loan-to-Income*), loan characteristics such as *Loan_Term*, and borrower characteristics, such as *Race* and *Age*. Column (1) shows a 3.4 % decrease in the difference in denial gap between minority and non-minority borrowers, down from a sample average denial gap of 16.3% (12.9% when accounting for fixed effects). Column (2) shows qualitatively similar results that are economically and statistically significant.

In addition, we examine differences in mortgage cost, proxied by the interest rates offered to minority borrowers compared to non-minority borrowers, as shown by the $DPL \times Minority$

interaction coefficient in Column (3)–(4). Column (3) shows an 8.2% decrease, indicating that DPL decreases minority interest rates by 8 bps. Column (4) includes further control variables for borrower characteristics; the coefficient remains negative and statistically significant, yielding economically and statistically similar results. Results using HMDA-GSE merged sample with credit scores in Panel B show qualitatively similar patterns. The effects on interest rates are still statistically significant, confirming the robustness of our findings.

We also use an alternative lending cost metric, the interest rate spread, which captures the difference between the loan’s annual percentage rate (APR) and the average prime offer rate (APOR) for a comparable transaction at the time. The results are shown in Table OA2 in the internet appendix; the coefficient of $DPL \times Minority$ indicates that DPL reduces the interest rate differential between minority and non-minority borrowers from the prime rate by 8.9 points. While minority borrowers generally face higher loan interest rates and denial rates compared to non-minorities, the reduction in loan costs and denial rates of minority borrowers is more pronounced than for non-minority borrowers.

Following Bertrand and Mullainathan (2003), we further capture dynamic effects by including dummy variables before and after DPL is signed into law in the following model¹²:

$$y_{i,l,t,c} = \sum_{k=-3}^3 \delta_k \times D_{t,c}^k \times Minority_i + \gamma Control_{i,t,c} + \varphi_l + \omega_t + \tau_c + \epsilon_{i,l,t,c} \quad (2)$$

where dummy variables $D_{t,c}^k$ capture periods preceding and following the DPL event, and δ_k represents the parameter of interest. We decompose the periods before and after the event into

¹² While recent methodological advances (e.g., Callaway & Sant’Anna, 2021) highlight potential biases in traditional TWFE estimators under staggered adoption, our design of restricting the control group to never-treated counties avoids any contamination from later-treated units. In our setting, our results are qualitatively similar when applying either approach (in line with discussions of Baker et al., 2022 on excluding treated units from the control group) and we further report stacked DiD estimators that present DPL effects for individual states, in Table A6 in the Internet Appendix.

seven bins: (i) if the mortgage application is submitted in a state three, two, or one year before the enactment of DPL, the respective indicator is equal to 1 (Pre_3 , Pre_2 , Pre_1); (ii) if the mortgage application is submitted in a state that enacts DPL in the same year, the indicator is equal to 1 ($Post_0$); (iii) if the mortgage application is submitted in a state one, two or three years after it has enacted DPL, the respective indicator is equal to 1 ($Post_1$, $Post_2$, $Post_3$). These indicators correspond to the three years before to the three years after the legislation's enactment year.

[Insert Figure 3 and Table 3 about here]

The results in Figure 3 and Table 3 indicate that the interaction terms on $Pre_3 \times Minority$, $Pre_2 \times Minority$, and $Pre_1 \times Minority$ related to minority applicants are statistically insignificant in both the interest rate and loan rejection regressions, thus supporting the validity of the parallel trends assumption. Conversely, the coefficients for $Post_0 \times Minority$, $Post_1 \times Minority$, $Post_2 \times Minority$ and $Post_3 \times Minority$ are mostly negative and statistically significant, consistent with our baseline results and demonstrating a sustained effect of DPL on interest and denial rates for minority applicants.

4.2 DPL Effects on Borrower Quality and Credit Demand

We examine whether the reduction in racial gaps after the enactment of DPL is driven by demand side effects and changes in borrower characteristics. Specifically, we investigate two dimensions: (i) whether borrower credit quality has changed following the introduction of privacy laws; (ii) whether effects are driven by an increase in borrower applications (due to increased demand or due to greater convenience of submitting applications).

First, we analyze whether applicants' ex-ante credit quality changed after the enactment of DPL, focusing on four key factors highly correlated with applicant credit scores and default

likelihood: income, loan-to-income (LTI) ratio, loan-to-value (LTV) ratio, and debt-to-income (DTI) ratio (Albanesi et al., 2022; Fuster et al., 2021). As shown in Figure OA4, the DTI distribution of applications by both minority and non-minority borrowers remains fairly stable, exhibiting a slight shift towards high DTI applicants for both minority and non-minority borrowers.

Following the baseline model, we included lender, county, and time-fixed effects and clustered at the state level but excluded any control variables, as shown in equation (3).

*Ex ante credit risk*_{*i,l,t,c*}

$$= \beta DPL_{ct} \times Minority_i + \theta DPL_{ct} + \mu Minority_i + \varphi_l + \omega_t + \delta_c + \epsilon_{i,l,t,c} \quad (3)$$

Additionally, to further consolidate the core variables that capture applicants' ex-ante credit risk (*Ex ante credit risk*_{*i,l,t,c*}) (i.e., the four key factors mentioned above), we employ principal component analysis (PCA) to construct a composite measure of credit quality. We generated two principal components, *PCA1* and *PCA2*, with *PCA1* explaining 42.51% of the total variance and *PCA1* and *PCA2* combined explaining a cumulative 68.78% of the variance. We created a new variable, *PCA1_2*, as a weighted linear combination of *PCA1* and *PCA2*, using their explained variance proportions as weights.

[Insert Table 4 about here]

Our results in Panel A of Table 4 show no significant changes in the credit quality of borrowers across different racial groups before and after the enactment of DPL, further alleviating concerns about changes in borrower characteristics following the enactment of DPL.

We further analyze the confounding effects of loan applications by borrowers. For this analysis, we aggregate data by county, lender, and year. These aggregated datasets record the total number of loan applications received and the total loan amounts by financial institutions within specific timeframes, thus revealing the overall dynamics of borrower demand in the loan market.

To assess the impact of DPL on overall mortgage demand and by minority borrowers, we estimate the following model (4):

$$y_{l,t,c} = \alpha + \beta DPL_{ct} + \gamma Control_{l,t,c} + \varphi_l + \omega_t + \tau_c + \epsilon_{l,t,c} \quad (4)$$

where l , t , and c represent the lender, year, and county, respectively. To explore how DPL influences loan demand across different types of borrowers, we have consolidated the data into two samples: one comprising all borrower applications within a specific county c , lender l , and year t ; and another for minority borrower applications only. The interaction variable DPL_{ct} represents lender effects in counties where DPL is enacted. Control variables (same as in the baseline model) are similarly aggregated at lender, county, and year level. We further include lender, year, and county fixed effects to account for time-varying impact across lenders and regions. We utilize three dependent variables $y_{l,t,c}$: *Applications*, which aggregates the number of mortgage loan applications for all applicants and from only minorities annually at lender and county levels; $Ln(Loan\ amounts)$, the natural logarithm of total mortgage loan amount requested from all applications and those from minorities; and *Approval rate* which aggregates the rates of loan approvals across all groups under the same conditions. Table 4, Panel B provides estimates of the impact of DPL on mortgage lending in terms of volume of applications (Columns (1)–(2)), total loan amount (Columns (3)–(4)), and approval rates for these applications (Columns (5)–(6)). The results in Panel B reveal that the enactment of DPL did not significantly increase the volume of mortgage applications or the total loan amounts submitted by all borrowers or minority applicants.

As the changing population of minority applicants could be another alternative driver for the reduction in racial disparity, we analyze potential changes in borrower income, loan amount, and credit risk. As more high-quality applicants choose to share data and apply for loans through FinTechs, low-quality applicants might suffer adverse effects from mortgage providers. To further

explore whether DPL impacts minorities' mortgage lending differently based on borrower types, we divide our samples into different groups: the borrowers whose income above (below) the median is designated as High (Low)-income borrowers, and those whose loan amount above (below) the median as High (Low)-loan amount borrowers. These subsample analyses are shown in Panels A and B of Table A3 in the Internet Appendix.

As shown in Panels A and B, we find no significant differences in effects based on borrowers' income or loan amount, suggesting that privacy regulations do not inherently differ in their impact across different income levels of minority borrowers and confirming uniform DPL effects. As the loan approval process increasingly relies on standardized processes and automation, it seems to also reduce human bias and differential treatment, so that applicants are judged primarily on characteristics pertaining to their credit risk regardless of income level, loan amount, or race.

4.3 Effects on Delinquencies

Given that we have shown that the implementation of DPL does not alter the underlying credit quality of applicants across racial groups (i.e., ex-ante credit risk remains unchanged) and that DPL reduces racial disparities in denial rates and interest rates, these findings indicate that the subsequent effects are not driven by changes in borrower composition.

To examine how DPL affects loan performance and to assess whether DPL not only improves access to credit of minority applicants but also evaluate implications for credit risks and the quality of credit allocation decisions, we use the merged HMDA–GSE dataset to analyze ex-post delinquency differences across racial groups. This dataset tracks the post-origination performance of individual loans and provides loan-level monthly updates, including delinquency status, applicant risk characteristics, and loan terms such as interest rates.

$$\begin{aligned}
& Delinquent_{i,l,t,c} \\
& = \beta DPL_{ct} \times Minority_i + \theta DPL_{ct} + \mu Minority_i + Control_{i,t,c} + \varphi_{lt} + \omega_{lc} \\
& + \epsilon_{i,l,t,c}
\end{aligned} \tag{5}$$

As shown in Equation (5), where i , l , t , and c denote the loan application, lender, year, and county, respectively. Our dependent variable $Delinquent_{i,l,t,c}$ is an indicator equal to 1 if the loan becomes more than 60-, 150-, or 180-days delinquent within the first one or two years after origination. In addition, we include direct measures of borrower risk from the GSE data, most notably the applicant's FICO credit score. The results are presented in Table 5.

[Insert Table 5 about here]

The results on delinquencies in Table 5 indicate that, following the enactment of data privacy legislation, the gap in mortgage default rates between minority and non-minority borrowers narrowed significantly. Specifically, the interaction term for DPL and minority status is negative and statistically significant across various measures of delinquency (60-, 150-, and 180-days past due), both within one and two years of loan origination. This suggests that despite a higher delinquency rate of minority borrowers compared to non-minority borrowers, minority borrowers experienced a greater reduction in default rates after DPL, converging towards that of non-minority borrowers. The findings imply that DPL led to fairer loan terms for minorities, such as lower interest rates, and thus a reduction in their monthly repayment burden. This might directly lower the possibility of them falling into financial distress and defaulting, with their default rate converging towards that of white borrowers. Overall, our results indicate that DPL does not change the inherent risks of minority borrowers for GSE loans but rather restrict the ability of lending institutions to impose additional risk premiums based on race, thereby improving the loan conditions for minorities and ultimately reducing their default rates.

5. Mechanisms

5.1 Underwriting Automation and Standardization

Automation and standardized credit assessment reduce the scope for subjective judgments human discretion, which historically tended to disadvantage minority borrowers. Automated systems rely more heavily on objective, codified information and less on idiosyncratic human assessments, thereby limiting channels through which implicit bias, stereotype-driven beliefs, or unequal scrutiny may influence lending decisions. Standardized credit models also enforce uniform evaluation rules and minimize heterogeneity in decision-making across loan officers, further contributing to a more consistent application of underwriting criteria across borrower groups.

To analyze whether automated underwriting and process standardization as the primary drivers of asymmetric DPL effects on minority and non-minority borrowers, we distinguish between loan decisions that rely on standardized assessment or not (based on information whether standard credit scoring models have been used) and usage of automated underwriting systems (AUS) during the underwriting process.

[Insert Table 6 about here]

The negative and significant coefficients for $DPL \times Minority \times AUS$ and $DPL \times Minority \times CS$ indicate that the reduction in denial rates and interest rate gaps between minority and non-minority borrowers after DPL is most pronounced for originations using automated underwriting systems and standardized credit models. While there is a larger racial gap in denial and interest rates for mortgage originations using AUS and standard credit models, the gap decreases significantly after DPL. In contrast, there does not seem to be an overall change in denial and interest rates for originations using AUS and CS.

5.2 Lender Type

We further explore the varying responses of different loan providers following DPL enactment, particularly concerning mortgages among minority groups. We follow the classification method of Buchak et al. (2018), distinguishing lenders into banks and non-bank financial institutions based on whether they also offer deposit services and further classify non-bank entities into FinTech companies based on whether their lending process is entirely online. To test the FinTech lending mechanism, we constructed dummy variables for FinTech firms (set to one if the lender neither accepts nor offers deposit service and the lender's loan process is entirely online, and zero otherwise). In addition, we apply a measure provided by Modi et al. (2022) that assessed financial institutions' IT investment levels by categorization of IT-related costs in their financial statements. To analyze the role of lender type, we include a four-way interaction effect to our model in Equation (1), estimating the following model (6):

$$y_{i,l,t,c} = \beta DPL_{ct} \times Minority_i \times FinTech_l + \theta DPL_{ct} + \mu Minority_i + \delta DPL_{ct} \times Minority_i + Control_{i,t,c} + \varphi_{lt} + \omega_{lc} + \epsilon_{i,l,t,c} \quad (6)$$

where i , l , t , and c represent the loan application, lender, year, and county, respectively. The variables are the same as in Equation (1). The variable *FinTech* indicates if the lending institution is a FinTech (*FinTech*=1). The coefficients of interest pertain to the sequence of estimates associated with the four-way interaction terms δ .

[Insert Table 7 about here]

In Panel A of Table 7, the significantly negative coefficient $DPL \times Minority \times Fintech$ indicates that, compared to traditional banks, FinTechs show a more pronounced effect in reducing racial disparities after the enactment of DPL, with FinTech lenders exhibiting an additional 2.7

percentage point reduction in minority denial gaps and a further 4.4 bps decline in interest-rate gaps relative to traditional lenders.

We further analyze differential effects of mortgage applications that are submitted to either FinTech or non-FinTech lenders. Following the borrower credit demand specification in Equation (3), we include an additional interaction variable, $DPL \times FinTech$, which indicates whether the mortgage application has been submitted to a FinTech lender. All other variables remain the same as in Equation 3.

Panel B of Table 7 details the effects of DPL on the volume of applications and distinguishes whether applications were submitted by minority borrowers and/or to a FinTech lender. In Column (2), we find that the coefficient on the interaction term $DPL \times Fintech$ is positive and significant, indicating that following the enactment of DPL, Fintech lenders experienced a significant increase in loan applications from minority groups. Concurrently, the legislation significantly increased the total amount of mortgage loans processed by FinTechs for all borrowers, including minorities, and improved approval rates for these applications.

In addition to differentiating between banks and FinTech lenders, we further conduct subsample analyses based on lenders' levels of IT infrastructure investment, allowing us to evaluate whether lenders with stronger versus weaker IT capacity respond differently to DPL. The subsample results for lenders with high and low levels of IT investments are shown in Table 8.

[Insert Table 8 about here]

Compared to Columns (1) and (2) of Panel B (where lending institutions have less IT investment), the coefficient of $DPL \times Minority$ shows greater effects in Columns (3) and (4) of Panel B (where lending institutions have higher IT investments), highlighting the important role of IT infrastructure and investment for mortgage origination processes.

To alleviate any concerns that these effects are driven by new lenders rather than existing lenders, we further classify lenders into existing and new lenders, demonstrating that the effects are driven by existing rather than new lenders (see results in the Internet Appendix, Table OA3).

5.3 Geographic Variation

We further assess whether effects are driven by the widening of financial services in areas with a high minority population and low bank branch density. Particularly in areas with a higher minority population, existing bank branch networks often provide insufficient services, and racial disparities are more salient (Wang and Zhang, 2020). We conducted subsample analysis based on the proportion of minority populations within census tracts and the density of bank branches within the county. We divide our samples into two groups: (i) those above the median are designated as high minority concentration areas (Minority intensity area), and those below the median as low minority concentration areas (Minority non-intensity area), based on the proportion of minority populations within a census tract¹³; (ii) based on the branch density of banks, with counties with branch density above the sample median classified as “Bank branch intensity area” and considered well-served by traditional lending, while those below the median were categorized as “Bank branch non-intensity area” and possibly underserved by traditional lending services. Results are presented in Table 9.

[Insert Table 9 about here]

The results show a wider offering of financial mortgages in high-minority intensity areas, reflected in the decrease in denial rates after the DPL enactment, particularly driven by FinTech

¹³ A census tract is a geographical area used by the United States Census Bureau for census and statistical analysis. Typically designed to be relatively permanent statistical subdivisions, each tract is intended to have a population size that is relatively balanced, usually around 4,000 people. The design of census tracts aims to maintain demographic and socio-economic stability over time, facilitating effective data analysis and community comparisons by governments, researchers, and the public.

lenders. Panel A in Table 9 indicates that DPL does not exhibit significant effects in areas with low minority concentration, while effects are significant in areas with higher concentrations of minorities (Columns (5)–(8)), confirming that DPL effects of reduced racial disparities in mortgage origination is more pronounced in areas with higher minority population. In contrast, areas with traditionally lower minority representation tend to exhibit lower baseline levels of racial gaps, limiting the marginal impact of DPL. These results suggest that DPL effects are stronger in regions with a higher concentration of minorities, enhancing financial inclusion in areas where racial gaps are more pronounced.

Moreover, areas with a high minority population have been shown to be traditionally underserved (Wang and Zhang, 2020), with a lower availability of banking services and bank branch density in these areas. However, the results are not fully explained by bank branch density. As shown in Panel B of Table 9, while we observe a significant reduction in denial rates and interest rates in areas with low bank branch density (Columns (5)–(8)), areas with higher density of bank branches still show a decrease in interest rates (Columns (3) and (4)), with the increasing of FinTech lending being one of the main drivers of reduction in racial disparities. Specifically, when we include an interaction term $DPL \times Minority \times FinTech$, we find that FinTechs exhibit a stronger mitigating effect on both denial rates and interest rate gaps between minority and non-minority applicants, regardless of the level of bank branch density. We confirm that DPL effects on racial gaps in mortgage lending are stronger in minority-dense areas and areas with sparse bank branch networks, thus widening access for underserved communities and minority borrowers.

6. Additional analysis

6.1 The Role of Racial Animus

To analyze the role of racial animus on the effects of DPL on racial disparities, we conduct subsample tests based on the geographical survey of racial bias and stereotypes, in particular whether the lending institution is located in a region with high/low implicit/explicit racial animus. Following previous studies (see Xu et al. (2014), Chernenko et al. (2024), Howell et al. (2024)) we construct county-level measures of “explicit bias” and “implicit bias” using average results from the Implicit Association Test (IAT).¹⁴ Within this framework, explicit racial animus refers to conscious preferences or prejudiced emotional responses based on race, and implicit racial animus refers to unconscious preferences or subtle discrimination in decision-making processes (Daumeyer et al., 2019).

We define explicit racial bias at the county level as strong preferences or aversions towards different races—for example, a strong preference for White individuals (indicating high bias against Black individuals), or a strong preference for Black individuals (indicating high bias against White individuals). A county is considered to have no explicit racial bias if it treats individuals of different races and skin colors equally, without regard to these characteristics (Chernenko et al., 2024; Schmidt et al., 2022). Figure OA5 displays the county-level distribution of these bias measures, each standardized to have zero-mean and unit variance. Subsample results are presented in Table 10.

¹⁴ We utilize the structure of the Implicit Association Test (IAT) to construct county-level “explicit bias” and “implicit bias” measures based on average responses to this survey, inferring the potential level of explicit and implicit bias of lenders in that county. Implicit biases are subconscious attitudes or stereotypes that influence perception and behavior. Figure A1 displays the distribution of county-level explicit and implicit bias measurements, each standardized to zero mean and unit variance. IAT asks respondents to rate their preference for Black versus White individuals on a scale from 1 to 7: 1 being “I greatly prefer African Americans over European Americans,” 4 being “I equally like European Americans and African Americans,” and 7 being “I greatly prefer European Americans over African Americans.” An absence of preference for one race over another would indicate a lack of explicit racial bias. Sources: <https://osf.io/gwofk/>.

[Insert Table 10 about here]

As illustrated in Table 10, Panel A, Columns (3) and (4), compared to counties without explicit racial bias, those with explicit racial bias witnessed a significant reduction in the disparities in rates and rejection rates between minority and non-minority applicants under DPL. In addition, we categorize counties in our sample into those with high and low implicit racial animus. As shown in Panel B of Table 10, we find that the outcomes were also sensitive to the level of high implicit racial animus at the county level. As indicated in Panel B, Columns (3) and (4), DPL had a uniformly significant effect on reducing rejection rate and interest rates for minorities, irrespective of the level of implicit racial animus in the county.

In addition, we conduct a subsample analysis based on whether the applicant's race is likely to be observable through facial or surname recognition. If lenders can infer the borrower's race from such subjective cues, this may affect how information is incorporated into mortgage approval decisions. Table OA4 presents these results. We find that the coefficient on $Treat \times Post \times Minority$ is larger for loan applications in which race is less observable (Columns (3) and (4)) compared with cases where race is more easily identified (Columns (1) and (2)). This suggests that DPL has a greater equalizing effect precisely when subjective racial cues are less available to lenders, consistent with the view that restricting access to identifying information reduces the scope for discretion-based or bias-driven decision making.

6.2 Differentiation by Loan Purpose

We further assess the impact of DPL on racial disparities in mortgage outcomes across different mortgage loan purposes, in particular home purchase and refinancing loans (we exclude home improvement loans). While refinancing loans cater to existing mortgage holders who are either

seeking to leverage their home equity or to change loan terms or duration, home purchase loans serve first-time buyers and those transitioning between homes.

In Table A2 of the Appendix, Columns (1)–(4) present the estimated results from the model in Equation (1), separately by purpose of the respective loan application (home purchase and refinancing). Each column includes only the sample corresponding to each loan purpose, and the analyses control for variables consistent with the baseline model. We find a significant reduction in racial disparities in both mortgage interest rates and rejection rates following the enactment of DPL, irrespective of the loan purpose. However, the comparison of coefficients of $DPL \times Minority$ across different columns shows that the reduction in the racial gap in interest rates is more pronounced for home purchase loans than for refinancing loans. We visually present the dynamic effects of DPL on the mortgage racial interest rate gap by loan purpose in Online Appendix Figure OA6. This discrepancy can be attributed to the fact that improvements in loan conditions in the refinancing market are more heavily dependent on the performance of existing loans and the value of the home rather than on initial credit assessments, making the rate adjustments for refinancing loans less sensitive compared to home purchase loans.

6.3 Heterogeneous Effects across Borrower Race and Ethnicity

To account for possible heterogeneity of effects depending on borrowers' race or ethnicity, we further differentiate between Black/non-Black and Hispanic/non-Hispanic borrowers, as presented in Table OA5. We confirm substantial gaps in denial and interest gaps for Black borrowers, as shown previously in related studies (such as Bartlett et al, 2022; Das et al., 2023) compared to non-Black and Hispanic borrowers. We further show that after DPL, the reduction in denial gaps is more pronounced for Black borrowers, compared to non-Black borrowers, while the reduction in

interest rate gap is more pronounced for Hispanic borrowers. As noted by Gerardi et al. (2023), racial gaps could be due to difficulties in obtaining refinancing mortgages by minority borrowers and the ability of non-Hispanic white borrowers to exploit periods of falling interest rates.

6.4 Robustness Checks

Following our baseline specification, we conduct various additional tests to validate the robustness of our findings, including: (i) differentiating by state-specific effects and DPL (Table OA6); (ii) examining the full sample of U.S. states and also focuses on neighboring counties during periods when the DPL is in effect (Table OA7); (iii) analyzing the lagged effects of DPL (Table OA8); (iv) excluding samples from states where the mortgage was signed within the same year of DPL enactment (Table OA9); (v) excluding samples from the Covid period, specifically 2020 to 2021 (Table OA10); (vi) excluding samples from the ten largest mortgage lenders (Table OA11); (vii) using propensity score matching (PSM) methodology (Table OA12); and (viii) employing different combinations of fixed effects (Table OA13).

To address any concerns regarding state-specific effects or the heterogeneous impact of DPL across states, we report the effects for each state individually (based on separate regressions of the model in Equation (1) for subsamples consisting of each state and its neighboring states). The classification of U.S. states and the results can be found in Table OA6. We indicate the political affiliation of each state based on the outcome of the last three elections in 2016, 2020, and 2024, and classify states based on whether their DPL emphasizes Customer Protection (CP) or Transparency (TP). Our state results highlight some heterogeneous effects for different states. We observe some increases in denial rate disparities for Utah and Delaware, which represent a relatively small fraction in terms of observations and population. We further note that states that

explicitly mention data transparency in their DPL exhibit stronger reductions in interest rate gap, whilst some states that do not mention transparency exhibit increased denial gaps, strengthening the notion that the decrease in racial gaps stems from increased data transparency requirements of state legislators.

In addition, we extend our analysis to include samples from all states with and without enacted privacy laws (illustrated in Figure 1a), rather than restricting the sample to counties along state borders, presented in Panel A of Table OA7. Instead of using the enactment dates of DPL, we focus on dates when DPL come into effect (as shown in Figure OA1), in contrast to the signing dates utilized in Sections 4.1 and 4.2. Consistent with baseline results, all models demonstrate a significantly negative coefficient for $DPL_Effective \times Minority$, indicating that DPL significantly reduces interest and rejection rates for minority applicants after DPL becomes effective.

We also examined lagged effects by analyzing the impact of DPL on mortgage outcomes one year after enactment in Table OA8, which allows us to capture any delayed effects of the law, showing consistent lagged effects.¹⁵ As the signing date of DPL can fall at the beginning, middle, or end of a year, we further exclude observations from the year of the law's enactment to avoid ambiguity regarding whether that year should be classified as pre- or post-legislation. The results in Tables OA9 alleviate concerns, showing that states affected by DPL experienced a significant reduction in racial disparities in mortgage denial rates and interest rates, while unaffected states did not exhibit such patterns.

To address potential concerns that effects are focusing on the COVID-19 period, we exclude the COVID-19 period from the data. As shown in Table OA10, the results continue to be robust and significant after excluding the COVID-19 period.

¹⁵ We re-estimate the remaining tables with the lagged treatment variable. The results are qualitatively similar and are available upon request.

To further ensure that our findings are not driven by large banks or financial institutions, and to address concerns about cross-state operations of large national banks, we rank lenders based on total mortgage application volume and loan amounts and exclude observations from the ten largest U.S. lenders, accounting for approximately 30% of the mortgage market. Although DPL imposes regulatory requirements on businesses¹⁶, nearly all U.S. lenders meet at least one of the legislative criteria. The results, as shown in Table OA11, remain consistent with the baseline findings.

To address the potential concern about borrower heterogeneity between states with and without DPL, we follow Di Maggio and Yao (2021) and employ propensity score matching (PSM) methodology. For each loan in a state with DPL, we identify the closest match among loans in states without DPL. The results are shown in the Online Appendix, Table OA12, and again show effects that are consistent with the baseline model.

Finally, our robustness checks incorporate a variety of fixed effects configurations, combining different sets of lender, county, and year fixed effects to evaluate the consistency of our results when not controlling for lender effects, time variations, or county-specific factors. As demonstrated in Table OA13, the results remain robust and statistically significant, reinforcing the conclusion that DPL significantly reduces racial disparities in the mortgage market.

7. Conclusion

Based on the staggered introduction of state-wide U.S. data privacy legislation, we demonstrate that DPL had asymmetric effects on minority and minority borrowers in the mortgage market. We show a reduction in racial disparity between minority and non-minority borrowers, with more

¹⁶ The regulations under the California Consumer Privacy Act (CCPA) apply to businesses that meet at least one of the following criteria: (i) annual gross revenue exceeding \$25 million; (ii) possession of personal information from 50,000 or more consumers, households, or devices; or (iii) earning more than half of their annual revenue from the sale of consumers' personal information. See: <https://www.oag.ca.gov/privacy/ccpa>.

pronounced effects in areas with greater racial animus, higher minority population, and lower bank branch density. Our results are primarily driven by the shift towards automated and standardized underwriting practices, triggered by increasing compliance and transparency requirements posed by DPL. FinTech lenders further contribute to the reduction of racial disparities by providing mortgages to minority borrowers in areas with a high-minority population, and by attracting more applications from minority borrowers. Applicants' observable credit risk does not change with DPL, but the narrowing delinquency gap among actual borrowers implies that DPL leads to less bias in borrower selection and assessment of borrower credit risk.

Our research contributes to the literature on data security and privacy protection and on the role of FinTech lending in the mortgage lending sector and holds practical implications for both financial institutions and regulatory bodies. While the goal of DPL is to alleviate consumer concerns about privacy and enhance data security controls, it can have further unintended effects for financial inclusion and asymmetric effects on different groups of borrowers. Our results provide new insights into the ongoing discussions on financial inclusion and racial disparities in the mortgage market, offering valuable perspectives on regulatory initiatives to widen financial inclusion and credit access for underrepresented groups.

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Figures and Tables

Figure 1. Coverage of DPL in U.S. states and neighboring states, and counties

These figures show the DPL signing dates across various states in the United States from 2018 to 2023. Different shades of blue represent the signing dates, with darker shades indicating earlier signing dates and lighter shades indicating later signing dates of DPL. In Figure 1a, neighboring states that did not sign the legislation are marked in gray, while non-neighboring states are marked in white. Figure 1b focusses on counties in states with enacted DPL and that are adjacent to states with legislation enactment, with counties in neighboring states marked in gray, while non-neighboring counties are marked in white.

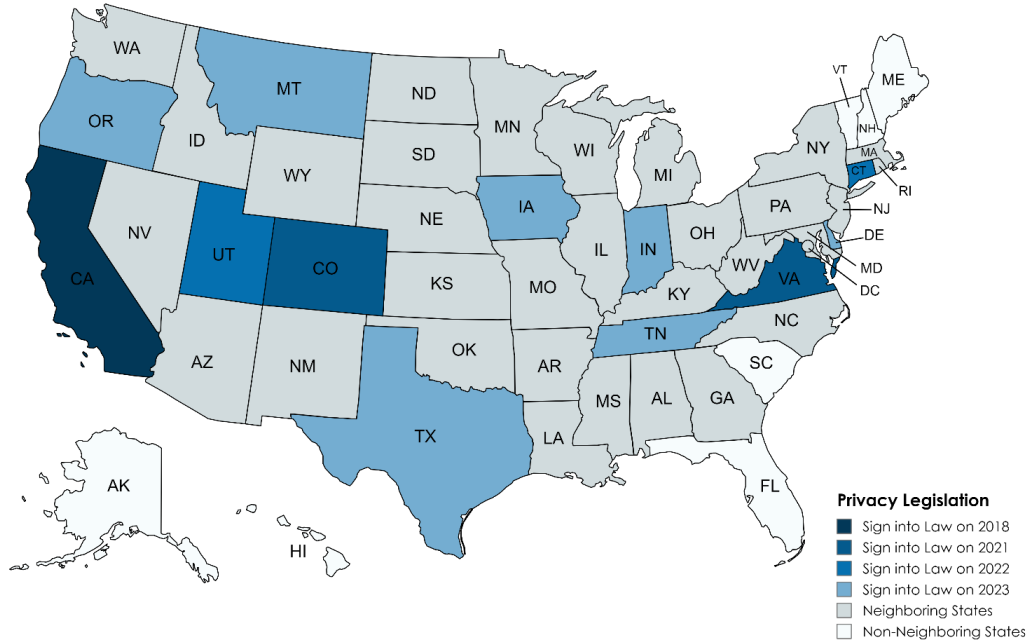


Figure 1a. State-level coverage and neighboring states

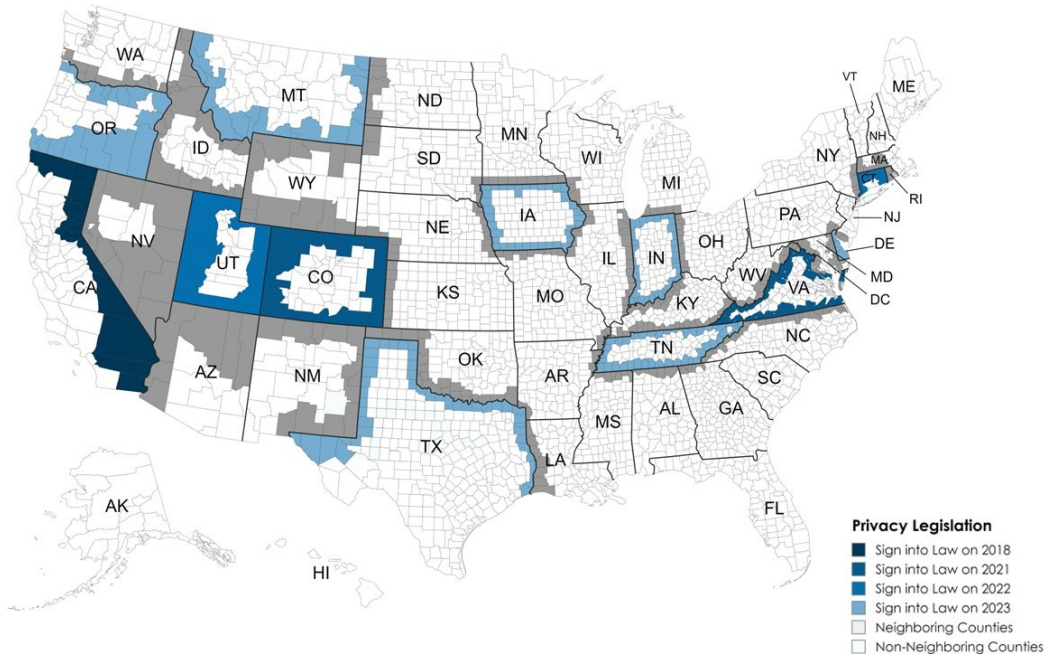


Figure 1b. County-level coverage and neighboring counties

Figure 2. Trends in Denial Rates and Interest Rate Spreads of Minorities and Non-Minorities

These figures show the average denial rates and interest rate spreads for minority and non-minority borrowers in the years around DPL enactment. The blue line represents the average outcomes for minority borrowers, while the red line depicts those for non-minority borrowers. The timelines across states are synchronized based on the respective years when the state adopted DPL. Dashed lines mark the year of enactment, distinguishing the periods before and after the legislation's adoption. Our analysis is confined to counties that enacted privacy protection laws within the sample period.

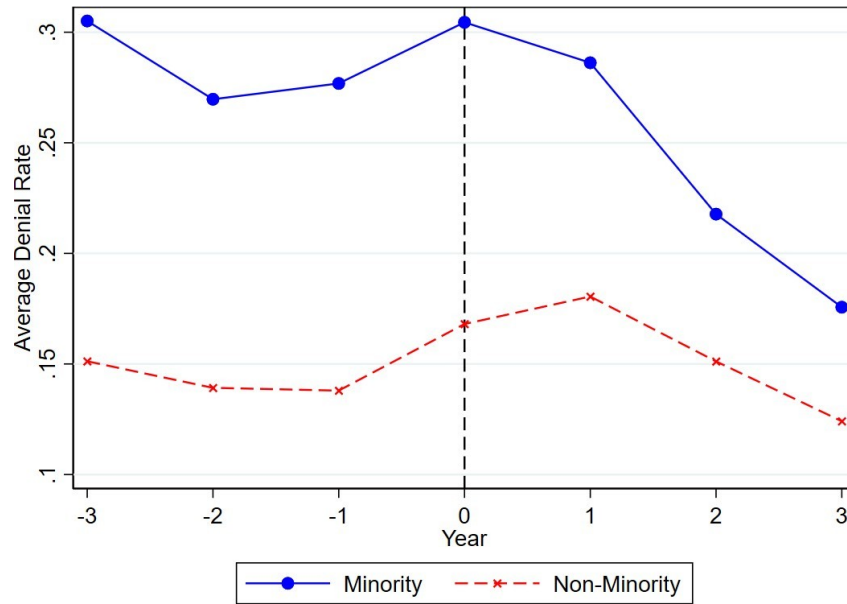


Figure 2a. Denial rates of minorities and non-minorities

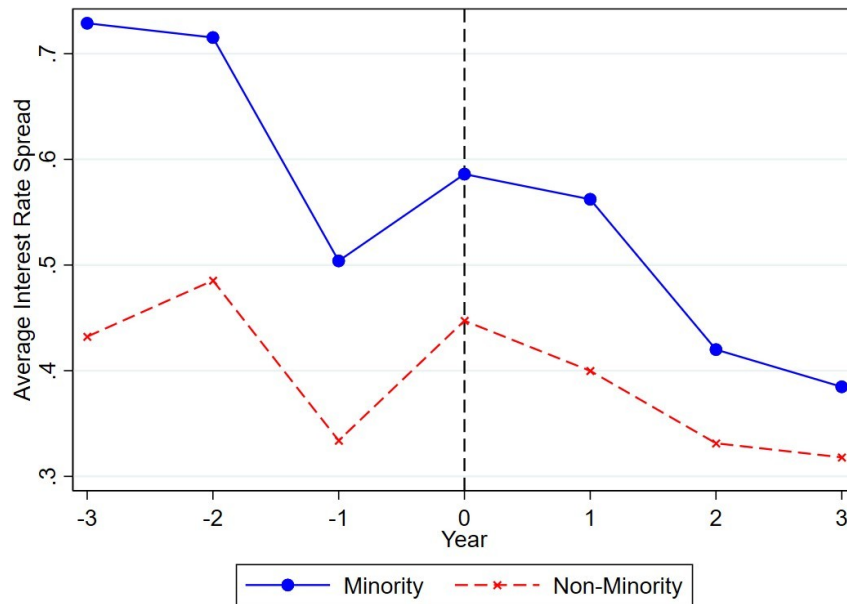


Figure 2b. Interest rate spreads of minorities and non-minorities

Figure 3. Dynamic Effects of DPL on Mortgage Racial Gap

These figures show the dynamic effects of DPL on interest rates and denial rates around the enactment date of the DPL. The horizontal axis presents the year relative to the year of the enactment of DPL, split into 3 years prior to enactment (*Pre_3*, *Pre_2*, *Pre_1*), 3 years after enactment (*Post_1*, *Post_2*, *Post_3*), and the same year *Post_0*. The vertical axis presents the difference in denial rate (Figure 3a) and interest rates (Figure 3b) of minority and non-minority borrowers, estimated using Model (2). The bar chart represents a 95% confidence interval.

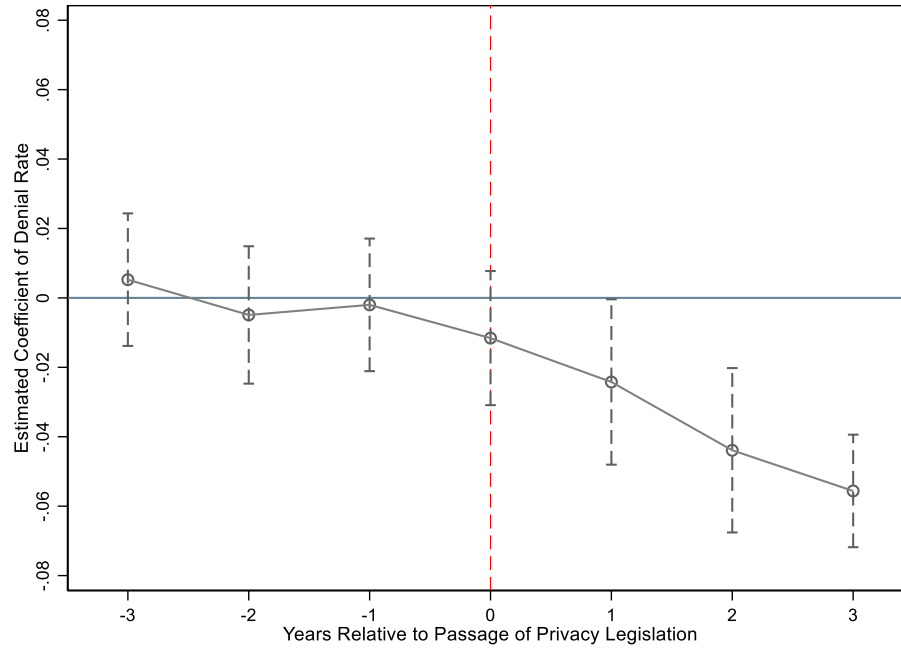


Figure 3a. Effect on denial rates

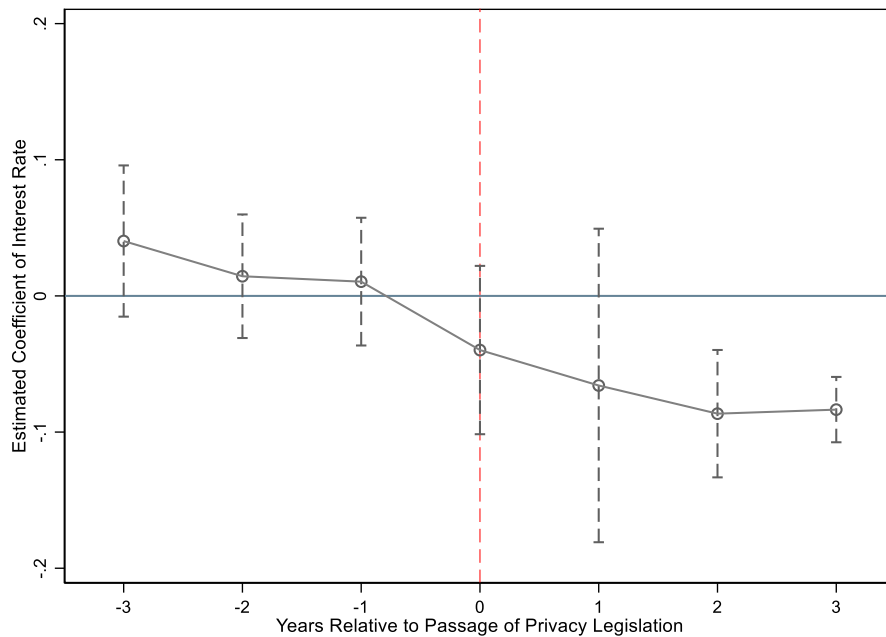


Figure 3b. Effect on interest rates

Table 1. Summary Statistics

The table reports descriptive statistics for all variables in our main empirical analysis for the full sample in Panel A and the subsamples of counties with and without DPL in Panel B. The sample consists of U.S. mortgage data from 2018 to 2023. The sample used for baseline results includes 7,222,834 application-level observations. For detailed information on the definition and construction of all variables, please refer to the Appendix Table OA1. Continuous variables, except macroeconomic variables, are winsorized at the first and 99th percentiles.

<i>Panel A: Full Sample</i>						
Variables	Obs.	Mean	Std. Dev.	1st Per.	Median	99th Per.
<i><u>Independent Variables</u></i>						
DPL	7,222,834	0.187	0.390	0	0	1
DPL_Effective	2,417,235	0.257	0.437	0	0	1
Minority	7,187,307	0.178	0.382	0	0	1
Female	7,222,834	0.257	0.437	0	0	1
<i><u>Dependent Variables</u></i>						
Denial	7,222,834	0.163	0.369	0	0	1
Interest Rate [%]	6,022,344	4.062	1.543	1.990	3.695	9.500
Interest Rate Spread	5,702,554	0.407	0.910	-1.820	0.281	4.144
<i><u>Control Variables</u></i>						
Credit Model	7,222,834	0.920	0.271	0	1	1
Debt_to_Income	7,222,834	0.620	0.486	0	1	1
LTV	7,222,834	0.646	0.261	0.055	0.714	1.000
Loan_to_Income	7,222,834	2.741	5.301	0.184	2.348	8.636
Loan_Amount [\$1,000]	7,222,834	265.32	202.82	15	225	1,075
Age category	7,222,834	3.792	1.430	1	4	7
Gender Observed	7,219,760	0.038	0.192	0	0	1
Race Observed	7,222,834	0.037	0.190	0	0	1
<i>Panel B: Counties with and without DPL</i>						
Variables	Non-Privacy counties		Privacy counties			
	Obs.	Mean	Obs.	Mean		
Minority	3,951,855	0.178	3,235,452	0.177		
Female	3,951,855	0.265	3,270,979	0.247		
Denial	3,951,855	0.165	3,270,979	0.161		
Interest Rate [%]	3,288,125	4.065	2,734,219	4.059		
Interest Rate Spread	3,121,184	0.398	2,581,370	0.418		
Credit Model	3,951,855	0.918	3,270,979	0.922		
Debt_to_Income	3,951,855	0.617	3,270,979	0.623		
LTV	3,951,855	0.650	3,270,979	0.642		
Loan_to_Income	3,951,855	2.675	3,270,979	2.821		
Loan_Amount [\$1,000]	3,951,855	263.65	3,270,979	267.34		
Race Observed	3,951,855	0.038	3,270,979	0.037		
Gender Observed	3,950,403	0.039	3,269,357	0.038		
Age	3,951,855	3.766	3,270,979	3.823		

Table 2. Effect of DPL on Mortgage Racial Denial and Interest Gap

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on denial rate and interest rate among minorities in the U.S. from 2018 to 2023. *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. The dependent variables are the dummy variable *Denial*, that represents whether this application has been rejected or not, and *Interest Rate*, that represents the cost of mortgages. Detailed information on all control variables is in Internet Appendix Table OA1. We include only mortgage application samples from *counties* in states where DPL has been enacted and counties adjacent to these states with DPL (see Figure 1b). Panel A presents the full HMDA application sample, while Panel B reports the results based on the HMDA–GSE merged data. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Full Sample Baseline</i>				
	Denial		Interest Rate	
	(1)	(2)	(3)	(4)
DPL	0.004*	0.003	0.021	0.020
	(1.91)	(1.60)	(1.28)	(1.22)
Minority	0.089***	0.083***	0.099***	0.103***
	(14.81)	(14.28)	(10.62)	(10.44)
DPL × Minority	-0.034***	-0.034***	-0.073***	-0.082***
	(-3.48)	(-3.45)	(-3.49)	(-4.04)
Credit Model		-0.203***		-0.641***
		(-15.27)		(-11.89)
Debt_to_Income		0.078***		0.067***
		(37.89)		(22.19)
LTV		-0.078***		-0.336***
		(-10.34)		(-3.82)
Loan_to_Income		0.005***		-0.010***
		(13.95)		(-6.64)
Loan_Term		0.000***		0.002***
		(7.16)		(12.75)
Race Observed		0.015***		0.130***
		(3.53)		(6.53)
Age		0.002*		-0.010*
		(1.72)		(-1.82)
Constant	0.145***	0.280***	4.045***	4.154***
	(136.80)	(20.87)	(1,086.52)	(35.18)
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	7,111,837	7,111,837	5,948,929	5,948,929
R-squared	0.197	0.226	0.679	0.694

<i>Panel B. HMDA-GSE Merged Sample</i>				
	Interest Rate		Spread	
	(1)	(2)	(3)	(4)
DPL	0.010 (0.48)	0.008 (0.49)	0.014 (1.32)	0.012* (1.86)
Minority	0.087*** (6.25)	0.039*** (4.31)	0.115*** (8.36)	0.071*** (6.38)
DPL× Minority	-0.048** (-2.64)	-0.026** (-2.34)	-0.051*** (-3.72)	-0.034*** (-2.89)
Credit Model		-0.006 (-0.27)		0.014 (0.62)
Debt_to_Income		0.046*** (9.90)		0.025*** (6.09)
LTV		0.208*** (12.83)		0.378*** (24.07)
Loan_to_Income		-0.044*** (-21.45)		-0.027*** (-21.44)
Loan_Term		0.003*** (85.24)		-0.000*** (-10.53)
Race Observed		0.034*** (3.79)		0.023** (2.55)
Age		0.019*** (16.38)		0.019*** (14.89)
Credit Score		-0.002*** (-48.22)		-0.002*** (-31.42)
Constant	3.602*** (826.05)	4.033*** (98.10)	0.274*** (118.55)	1.970*** (44.18)
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	157,645	157,645	150,922	150,922
R-squared	0.767	0.813	0.283	0.360

Table 3. Dynamic Effects of DPL

The table presents the results of the dynamic regression model examining the effects of DPL on the interest rate and denial rates for minority mortgage borrowers. The analysis replaces the test variable *DPL* with a set of dummy variables *Pre_3*, *Pre_2*, *Pre_1*, *Post_0*, *Post_1*, *Post_2*, and *Post_3*, which capture the periods before and after the DPL. T-statistics are presented in parentheses. The coefficients on all the *Pre_3* to *Post_3* are omitted for brevity. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Denial	Interest Rate
	(1)	(2)
<i>Pre_3</i> × Minority	0.005 (0.55)	0.040 (1.46)
<i>Pre_2</i> × Minority	-0.005 (-0.50)	0.014 (0.64)
<i>Pre_1</i> × Minority	-0.002 (-0.21)	0.010 (0.45)
<i>Post_0</i> × Minority	-0.012 (-1.21)	-0.040 (-1.30)
<i>Post_1</i> × Minority	-0.024** (-2.05)	-0.066 (-1.15)
<i>Post_2</i> × Minority	-0.044*** (-3.74)	-0.086*** (-3.73)
<i>Post_3</i> × Minority	-0.056*** (-6.92)	-0.084*** (-7.02)
Minority	0.092*** (11.52)	0.099*** (8.43)
Pre and Post	Y	Y
Lender FE	Y	Y
Year FE	Y	Y
County FE	Y	Y
Observations	7,187,098	5,993,673
R-squared	0.174	0.639

Table 4. DPL Effects on Borrower Credit Risk and Credit Demand

This table examines DPL effects on borrowers' credit quality and loan applications. *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. In Panel A, we focus on four key variables correlated with credit scores and default risk: income, loan-to-income ratio, loan-to-value ratio, and debt-to-income ratio. Using principal component analysis (PCA), we construct two components (PCA_1 and PCA_2) and combine them into a weighted linear index labeled PCA1_2, used as an alternative measure of credit risk. The dependent variables in Columns (1) – (3) are *Loan_to_Income*, *PCA_1*, and *PCA1_2*, respectively, to capture applicant credit risk. In Panel B, we employed three primary dependent variables: *Applications*, which aggregates the number of mortgage applications by county, lender, and year for all applicants, and *Minority %* (i.e., the ratio of minority borrowers to the total number of applicants. *Ln (Loan Amount)*, the natural logarithm of the total mortgage amount requested by all applicants, *Minority %* (i.e. the ratio of the total loan amount of minority borrowers to the total loan amount of all applicants) under identical conditions; and *Approval Rate*, the aggregate loan approval rates for all groups under the same conditions. T-statistics are shown in parentheses. Standard errors are clustered at the state level. We employ fixed effects at the lender, year, and county levels. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Effects on Borrower Credit Risk</i>						
	Loan_to_Income		PCA_1		PCA1_2	
	(1)		(2)		(3)	
DPL	-0.095*		-0.018		-0.012	
	(-1.77)		(-0.99)		(-0.99)	
Minority	0.146***		-0.135***		-0.090***	
	-4.07		(-4.38)		(-4.38)	
DPL × Minority	0.019		0.01		0.006	
	-0.45		-0.31		-0.31	
Lender FE	Y		Y		Y	
Year FE	Y		Y		Y	
County FE	Y		Y		Y	
Observations	7187098		7187098		7187098	
R-squared	0.053		0.208		0.208	
<i>Panel B. Demand side analysis of all lenders</i>						
	Applications		Ln (Loan amounts)		Approval rate	
	All	Minority %	All	Minority %	All	Minority
	(1)	(2)	(3)	(4)	(5)	(6)
DPL	3.52	-0.001	0.008	0.029	-0.001	0
	-1.57	(-0.53)	-0.31	-0.45	(-0.26)	(-0.10)
Controls	Y	Y	Y	Y	Y	Y
Lender FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
County FE	Y	Y	Y	Y	Y	Y
Observations	318,392	318,392	318,392	318,392	318,392	318,151
R-squared	0.099	0.296	0.504	0.331	0.335	0.244

Table 5. DPL Effects on Mortgage Delinquencies

These tables report the effects after DPL on several measures of delinquency, defined as loans becoming 60, 150, or 180+ days past due within one or two years of origination. For each delinquency horizon, the table shows the coefficients for DPL, minority status, and their interaction. The control variables include *Credit Model*, *Debt_to_Income*, *Loan_to_Value*, *Loan_to_Income*, *Loan_Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix Table OA1. T-statistics are shown in parentheses. Standard errors are clustered at the state level. We employ fixed effects at the year $\times$ lender, and county $\times$ lender levels. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Loan performance within one year of origination</i>			
	Delinquent 60	Delinquent 150	Delinquent 180
	(1)	(2)	(3)
DPL	0.002 (0.48)	0.001 (0.56)	0.001 (0.54)
Minority	0.006*** (3.19)	0.003*** (3.26)	0.001** (2.06)
DPL $\times$ Minority	-0.008*** (-2.81)	-0.003** (-2.58)	-0.002*** (-3.18)
Credit Score	-0.000*** (-20.27)	-0.000*** (-9.33)	-0.000*** (-8.67)
Control Variables	Y	Y	Y
Year $\times$ Lender FE	Y	Y	Y
County $\times$ Lender FE	Y	Y	Y
Observations	117,329	117,329	117,329
R-squared	0.110	0.078	0.070
<i>Panel A. Loan performance within two years of origination</i>			
	Delinquent 60	Delinquent 150	Delinquent 180
	(1)	(2)	(3)
DPL	0.001 (0.22)	0.005* (2.01)	0.003* (1.72)
Minority	0.017*** (5.56)	0.006*** (4.37)	0.004*** (2.95)
DPL $\times$ Minority	-0.010** (-2.54)	-0.003* (-1.83)	-0.003** (-2.66)
Credit Score	-0.001*** (-27.04)	-0.000*** (-18.39)	-0.000*** (-12.71)
Control Variables	Y	Y	Y
Year $\times$ Lender FE	Y	Y	Y
County $\times$ Lender FE	Y	Y	Y
Observations	117,329	117,329	117,329
R-squared	0.125	0.089	0.081

Table 6. Role of Automation and Standardization in Mortgage Originations

These tables examine the subsample analysis of DPL effects on mortgage interest rates and denial rates for origination using automated underwriting systems (AUS) and standard credit models (CS). *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we use *Denial* to represent whether this application has been rejected or not and *Interest Rate* to represent the cost of mortgage loans. The control variables include *Credit Model*, *Debt_to_Income*, *Loan_to_Value*, *Loan_to_Income*, *Loan_Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables, see Internet Appendix Table OA1. T-statistics are shown in parentheses. Standard errors are clustered at the state level. We employ fixed effects at the year $\times$ lender, and county $\times$ lender levels. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
AUS	-0.241*** (-24.66)	-0.594*** (-12.24)		
Minority $\times$ AUS	0.037*** (5.31)	0.089*** (5.21)		
DPL $\times$ AUS	0.000 (0.04)	0.010 (0.22)		
DPL $\times$ Minority $\times$ AUS	-0.018** (-2.48)	-0.045** (-2.64)		
CS			-0.214*** (-15.18)	-0.654*** (-12.19)
Minority $\times$ CS			0.073*** (13.55)	0.096*** (9.72)
DPL $\times$ CS			-0.001 (-0.31)	0.027 (1.07)
DPL $\times$ Minority $\times$ CS			-0.030*** (-3.25)	-0.076*** (-3.96)
Control Variables	Y	Y	Y	Y
Year $\times$ Lender FE	Y	Y	Y	Y
County $\times$ Lender FE	Y	Y	Y	Y
Observations	7,111,837	5,948,929	7,111,837	5,948,929
R-squared	0.248	0.701	0.224	0.694

Table 7. Role of Fintech Lenders

This table presents the results for the role of Fintech. We include a *FinTech* dummy variable indicating whether the lender is a FinTech (see classification in Section 3.2). *DPL* indicates whether the mortgage application was submitted in a state where DPL was enacted. Panel A presents the supply-side analysis of FinTech lenders, while Panel B reports the demand-side analysis. In Panel A, Column (1) displays the interaction effects on denial rates, whereas Columns (2) and (3) present the effects on interest rates, with Column (2) based on the full HMDA sample and Column (3) on the HMDA–GSE merged sample. In Panel B, we employ three primary dependent variables. *Applications*, *Ln(Loan Amount)* and *Approval Rate* (consistent with Table 4). The control variables include *Credit Model*, *Debt to Income*, *Loan to Value*, *Loan to Income*, *Loan Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables, see Internet Appendix Table OA1. T-statistics are shown in parentheses. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Supply side analysis of Fintech lenders</i>						
	Denial		Interest Rate			
	(1)		(2)		(3)	
DPL	0.003		0.019		0.005	
	(1.66)		(1.16)		(0.27)	
Minority	0.083***		0.104***		0.040***	
	(14.71)		(10.73)		(4.49)	
DPL× Minority	-0.030***		-0.072***		-0.022**	
	(-3.09)		(-3.91)		(-2.23)	
DPL× Minority× Fintech	-0.027***		-0.044**		-0.039***	
	(-9.32)		(-2.53)		(-4.67)	
Credit Score					-0.002***	
					(-49.68)	
Control Variables	Y		Y		Y	
Year × Lender FE	Y		Y		Y	
County × Lender FE	Y		Y		Y	
Observations	6,877,345		5,738,176		151,127	
R-squared	0.227		0.694		0.813	
<i>Panel B. Demand side analysis of Fintech lenders</i>						
	Applications		Ln (Loan amounts)		Approval rate	
	All	Minority %	All	Minority %	All	Minority
	(1)	(2)	(3)	(4)	(5)	(6)
DPL	3.198	-0.003	-0.009	-0.019	-0.002	-0.001
	(1.36)	(-1.05)	(-0.36)	(-0.27)	(-0.95)	(-0.62)
DPL × Fintech	3.025	0.007**	0.151***	0.386***	0.009**	0.007**
	(1.31)	(2.36)	(2.83)	(2.99)	(2.65)	(2.52)
Controls	Y	Y	Y	Y	Y	Y
Lender, year, county FE	Y	Y	Y	Y	Y	Y
Observations	306,690	306,690	306,690	306,690	306,690	306,463
R-squared	0.098	0.295	0.503	0.329	0.339	0.242

Table 8. Role of Bank Characteristics

The subsample analyses explore heterogeneity in the effects of the DPL. Panel A differentiates applications based on whether it was submitted to a lender with high levels of IT investment, while Panel B distinguishes states by whether their bank branch intensity is above or below the sample median. *DPL* indicates whether the mortgage application was submitted in a state where DPL was enacted, and *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we use *Denial* to represent whether this application has been rejected or not and *Interest Rate* to represent the cost of mortgage loans. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix Table OA1. T-statistics are shown in parentheses. Standard errors are clustered at the state level. We employ fixed effects at the year $\times$ lender, and county $\times$ lender levels. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. IT investment</i>				
	Low IT investment		High IT investment	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	-0.004*	0.023	0.004	0.017
	(-1.72)	(1.21)	(1.05)	(0.54)
Minority	0.114***	0.102***	0.123***	0.219***
	(15.86)	(7.27)	(23.60)	(14.30)
DPL $\times$ Minority	-0.015	-0.067	-0.041***	-0.095***
	(-1.60)	(-1.58)	(-4.58)	(-3.26)
Control Variables	Y	Y	Y	Y
Year $\times$ Lender FE	Y	Y	Y	Y
County $\times$ Lender FE	Y	Y	Y	Y
Observations	1,258,213	1,016,018	1,285,477	1,013,365
R-squared	0.242	0.681	0.157	0.681
<i>Panel B. Bank Branch Intensity</i>				
VARIABLES	High Branch Intensity		Low Branch Intensity	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	-0.004	-0.108**	0.004	0.031*
	(-1.63)	(-2.77)	(1.58)	(2.00)
Minority	0.095***	0.056***	0.075***	0.029***
	(27.91)	(3.85)	(11.40)	(5.43)
DPL $\times$ Minority	-0.011	0.002	-0.027***	-0.018**
	(-1.42)	(0.06)	(-2.81)	(-2.34)
Control Variables	Y	Y	Y	Y
Year $\times$ Lender FE	Y	Y	Y	Y
County $\times$ Lender FE	Y	Y	Y	Y
Observations	2,356,686	48,266	4,754,272	109,079
R-squared	0.234	0.832	0.222	0.806

Table 9. Geographic Variation for Areas with Minority and Bank Branch Density

These tables examine the geographic variation of how DPL affects mortgage interest rates and denial rates. *DPL* indicates whether the mortgage application was submitted in a state where DPL was signed. Panel A reports results for applications in tracts with low and high minority population. Panel B reports results for applications in states with high and low intensity of bank branches. The coefficients for control variables are omitted for brevity and are consistent with baseline models. T-statistics are shown in parentheses. Standard errors are clustered at the state level. We employ fixed effects at the year $\times$ lender, and county $\times$ lender levels. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Minority intensity subsample analysis</i>				
	Minority low-intensity area		Minority high-intensity area	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	0.003	-0.012	0.002	0.035
	(1.43)	(-0.79)	(0.67)	(1.54)
Minority	0.079***	0.093***	0.085***	0.111***
	(20.65)	(7.88)	(11.50)	(10.06)
DPL $\times$ Minority	-0.008	-0.025	-0.036***	-0.086***
	(-1.44)	(-0.62)	(-4.16)	(-6.08)
DPL $\times$ Minority $\times$ Fintech	-0.027**	-0.005	-0.026***	-0.054***
	(-2.44)	(-0.30)	(-13.39)	(-4.70)
Control Variables	Y	Y	Y	Y
Year $\times$ Lender FE	Y	Y	Y	Y
County $\times$ Lender FE	Y	Y	Y	Y
Observations	3,017,030	2,549,872	3,639,032	3,002,297
R-squared	0.236	0.709	0.225	0.707
<i>Panel B. Bank branch intensity subsample analysis</i>				
	High branch intensity area		Low branch intensity area	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	0.004	0.020	-0.003	0.003
	(1.68)	(1.04)	(-1.36)	(0.08)
Minority	0.075***	0.099***	0.096***	0.111***
	(11.82)	(6.97)	(27.02)	(10.49)
DPL $\times$ Minority	-0.023**	-0.065***	-0.009	-0.215***
	(-2.45)	(-3.19)	(-1.10)	(-8.36)
DPL $\times$ Minority $\times$ Fintech	-0.026***	-0.046***	-0.029***	-0.111**
	(-8.59)	(-2.95)	(-2.98)	(-2.24)
Control Variables	Y	Y	Y	Y
Year $\times$ Lender FE	Y	Y	Y	Y
County $\times$ Lender FE	Y	Y	Y	Y
Observations	4,578,447	3,807,497	2,298,048	1,929,796
R-squared	0.224	0.690	0.236	0.705

Table 10. Role of Racial Animus

These tables examine DPL effects across areas with high and low levels of racial animus. *DPL* indicates whether the mortgage application was submitted in a state where DPL was enacted. Panel A and Panel B show subsample results based on the benchmark model in Equation (1) categorized by subsamples of mortgage applications. Panel A shows results for applications in counties with low overt explicit racial biases (Columns (1) and (2)) and those with high overt racial biases (Columns (3) and (4)), illustrating the effects of explicit racial bias. Panel B reports results for applications in counties with low covert implicit racial biases (Columns (1) and (2)) and those with high covert racial biases (Columns (3) and (4)), reflecting the impact of implicit racial animus. The coefficients for control variables are omitted for brevity. T-statistics are shown in parentheses. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Explicit Bias classification</i>				
	Low Animus		High Animus	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	0.004*** (2.92)	-0.001 (-0.07)	0.006*** (3.69)	0.033 (1.05)
Minority	0.087*** (14.14)	0.119*** (6.77)	0.082*** (11.40)	0.099*** (8.77)
DPL × Minority	-0.017*** (-3.22)	-0.035 (-0.86)	-0.039*** (-4.63)	-0.091*** (-7.67)
Control Variables	Y	Y	Y	Y
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	1,994,783	1,670,792	5,116,023	4,277,071
R-squared	0.245	0.720	0.219	0.685
<i>Panel B. Implicit Bias classification</i>				
	Low Animus		High Animus	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	-0.000 (-0.11)	0.018 (1.26)	0.001 (0.46)	0.014 (0.41)
Minority	0.077*** (9.08)	0.094*** (5.30)	0.089*** (15.37)	0.112*** (13.22)
DPL × Minority	-0.005 (-0.60)	-0.025 (-0.63)	-0.049*** (-8.23)	-0.107*** (-10.82)
Control Variables	Y	Y	Y	Y
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	3,225,977	2,688,718	3,884,887	3,259,225
R-squared	0.231	0.688	0.223	0.702

Appendix

Table A1. Details of State-level Data Privacy Legislation

	Event date	State	Description
(1)	Jun 28, 2018	California	The California Consumer Privacy Act (CCPA) was signed into law on June 28, 2018, and went into effect on January 1, 2020. The CCPA applies to any business, including any for-profit entity that collects consumers' data, that does business in California
(2)	Jul 7, 2021	Colorado	Colorado Senate Bill 190 (SB 190), also known as the Colorado Privacy Act (CPA), was introduced and signed into law on July 7, 2021, and its effective Date is July 1, 2023
(3)	May 10, 2022	Connecticut	Connecticut Data Privacy Act (CTDPA) has been signed into Law by Governor Ned Lamont on May 10, 2022, and its effective Date is July 1, 2023.
(4)	Mar 24, 2022	Utah	Utah Senate Bill 227 (SB 227), also known as the Utah Consumer Privacy Act (UCPA), was signed into law by Governor Spencer Cox on March 24, 2022. The law took effect on December 31, 2023.
(5)	Mar 2, 2021	Virginia	Virginia Senate Bill 1392 (SB 1392), known as the Virginia Consumer Data Protection Act (VCDPA), was signed into law by Governor Ralph Northam on March 2, 2021. The law took effect on January 1, 2023.
(6)	Sep 11, 2023	Delaware	The Delaware Personal Data Privacy Act (DPDPA) was signed into law on September 11, 2023, and it took effect on January 1, 2025.
(7)	May 1, 2023	Indiana	The Indiana Consumer Data Protection Act (ICDPA) was signed into law on May 1, 2023, and it took effect on January 1, 2026.
(8)	Mar 28, 2023	Iowa	Iowa Consumer Data Protection Act (ICDPA) was signed into law on March 28, 2023, and it took effect on January 1, 2025.
(9)	May 19, 2023	Montana	Montana Consumer Data Privacy Act (MCDPA) was signed into law on May 19, 2023, and it took effect on October 1, 2024.
(10)	Jun 22, 2023	Oregon	Oregon Consumer Privacy Act (OCPA) was signed into law on June 22, 2023, and it will take effect on July 1, 2024.
(11)	May 11, 2023	Tennessee	Tennessee Information Protection Act (TIPA) was signed into law on May 11, 2023, and it took effect on July 1, 2025.
(12)	Jun 18, 2023	Texas	Texas Data Privacy and Security Act (TDPSA) was signed into law on June 18, 2023, it took effect on July 1, 2024.

Table A2. Loan Purpose Differentiations

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on the denial rate and interest rate among minorities in the United States from 2018 to 2023, categorized by different loan purposes, i.e., Home purchase, Refinancing (including Cash-out refinancing). *DPL* indicates whether the mortgage application was submitted in a state where DPL was enacted, and *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we primarily use *Denial* to represent whether this application has been rejected or not, and *Interest Rate* to represent the cost of mortgage loans. The control variables include *Credit Model*, *Debt_to_Income*, *Loan_to_Value*, *Loan_to_Income*, *Loan_Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables, see Internet Appendix Table OA1. Continuous variables, except macroeconomic variables, are winsorized at the first and 99th percentiles. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

VARIABLES	Home Purchase		Refinancing	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	0.001 (0.87)	0.034* (1.86)	-0.002 (-0.07)	0.005* (1.70)
Minority	0.052*** (13.20)	0.057*** (4.48)	0.117*** (8.27)	0.086*** (11.85)
DPL × Minority	-0.021*** (-2.96)	-0.075*** (-3.45)	-0.071*** (-4.17)	-0.044*** (-3.83)
Control Variables	Y	Y	Y	Y
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	2,922,324	2,580,173	2,969,023	3,528,857
R-squared	0.302	0.726	0.678	0.181

Table A3. Borrower characteristics

This table presents the effects of DPL based on subsamples for borrowers with high and low income (Panel A) and borrowers with high and low loan amounts (Panel B), based on the regression model in Equation (1). The coefficients for control variables are omitted for brevity. For the definitions of all the control variables, see Internet Appendix Table OA1. T-statistics are shown in parentheses. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Borrower income subsample analysis</i>				
	High-income borrowers		Low-income borrowers	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	0.002 (0.90)	0.018 (1.30)	0.004 (1.36)	0.028 (1.53)
Minority	0.074*** (17.52)	0.148*** (19.86)	0.072*** (12.78)	0.070*** (8.29)
DPL × Minority	-0.033*** (-4.22)	-0.067*** (-2.89)	-0.030*** (-2.95)	-0.072*** (-5.34)
Control Variables	Y	Y	Y	Y
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	3,573,203	3,167,244	3,512,889	2,757,157
R-squared	0.166	0.714	0.266	0.689
<i>Panel B. Borrower loan amount subsample analysis</i>				
	High-loan amount borrowers		Low-loan amount borrowers	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	0.001 (0.58)	0.009 (0.68)	0.001 (0.29)	0.009 (0.29)
Minority	0.051*** (16.31)	0.072*** (7.75)	0.091*** (15.56)	0.075*** (5.99)
DPL × Minority	-0.018*** (-3.61)	-0.040*** (-2.85)	-0.029** (-2.35)	-0.103*** (-3.43)
Control Variables	Y	Y	Y	Y
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	3,627,791	3,246,440	3,627,791	2,679,414
R-squared	0.152	0.770	0.152	0.676

Online Appendix

Racial Disparities in the U.S. Mortgage Market: Evidence from Data Privacy Legislation

In this online appendix, we provide additional discussions and robustness tests that supplement the empirical analysis in the main body of the manuscript. Below is the list of contents:

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Figure OA2. Google Search Trends about DPL

Figure OA3. CFPB complaints

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Table OA9. Robustness_ Exclude Privacy legislation current year

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Table OA13. Robustness_ Different combinations of Fixed effects

Table OA14. Denial Reason Differentiations

Discussion of Loan Denial Reasons

We explore whether DPL leads to differential denial reasons for mortgage applications. Utilizing the reasons for loan denials among minority groups, we assess the denial reasons for minority borrowers due to insufficient data. The HMDA provides detailed classifications of primary denial reasons offered by lending institutions, which include: (i) issues related to credit history, such as insufficient credit references, unacceptable credit reference types, no credit file, limited credit experience, poor credit performance, delinquencies on current or past obligations, frequent inquiries by credit agencies, foreclosures, repossessions, write-offs, collection actions, judgments, and bankruptcy; (ii) insufficient cash for downpayments and closing costs; (iii) inadequate debt-to-income ratios where the applicant's income fails to cover the requested amount or excessive obligations related to income do not meet credit standards; (iv) employment history concerns like temporary or informal employment and insufficient length of employment; (v) inadequate value or type of collateral, and (vi) denied mortgage insurance.

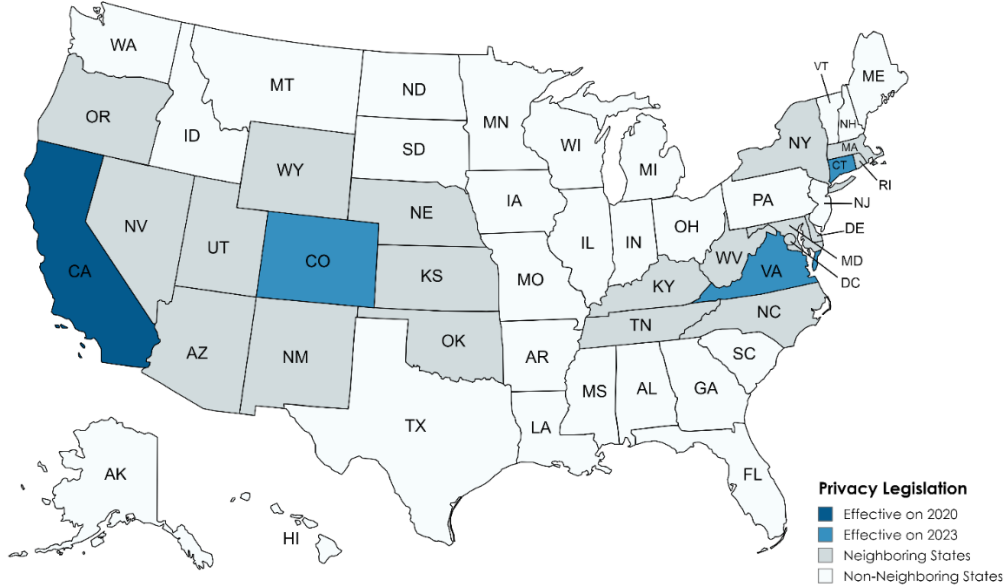
As shown in Table OA14, the sample is exclusively on observations of mortgage loan applications that were denied and for which the lending institutions disclosed the reasons for denial. The dependent variables in Columns (1) to (6) correspond to different reasons for loan denials: Credit history (equals 1 if the mortgage loan was denied due to insufficient credit history, otherwise 0), Insufficient cash (equals 1 if the mortgage loan was denied due to insufficient cash for downpayment and closing costs, otherwise 0), Debt-to-income ratio (equals to 1 if the mortgage loan was denied because the debt-to-income ratio did not meet credit requirements, otherwise 0), Employment history (equals to 1 if the mortgage loan was denied due to inadequate employment history or duration of employment not meeting credit requirements, otherwise 0), Collateral (equals to 1 if the mortgage loan was denied due to inadequate value or type of collateral, otherwise

0), and Mortgage insurance denied (equals to 1 if the mortgage loan was denied due to reasons related to mortgage insurance, otherwise 0).

[Insert Table OA14 about here]

In Table OA14, Column (1), the coefficient for $DPL \times Minority$ is -0.016, indicating that DPL has reduced the likelihood of minorities being denied loans due to insufficient credit history. The introduction of privacy laws has enhanced transparency and accountability in lending, allowing financial institutions to facilitate more efficient and complete data sharing and to perform more accurate risk assessments for borrowers traditionally lacking sufficient credit history, thus reducing denials due to insufficient credit references or the absence of a credit file. Additionally, as shown in Column (2) of Table OA14, there is relatively weak evidence that DPL may decrease the likelihood of minorities being denied loans due to insufficient funds (e.g., down-payments, closing costs). However, we did not find evidence suggesting a reduction in denials due to Debt-to-income ratio, Employment history, Collateral, or Mortgage insurance denied.

This figure shows the DPL effective dates across various states in the United States from 2018 to 2023. Different shades of blue represent the effective dates, with darker shades indicating earlier effective dates. Neighboring states that do not have DPL in effect are marked in gray, while non-neighboring states are marked in white.



These figures illustrate the Google search interest in the terms "California Consumer Privacy Act" and "Consumer Privacy" from May 2015 to 2024. The y-axis represents the normalized search interest (ranging from 0 to 100), and the x-axis indicates the timeline in monthly intervals. The figure shows fluctuations in search interest, with multiple peaks and troughs across the observed period.

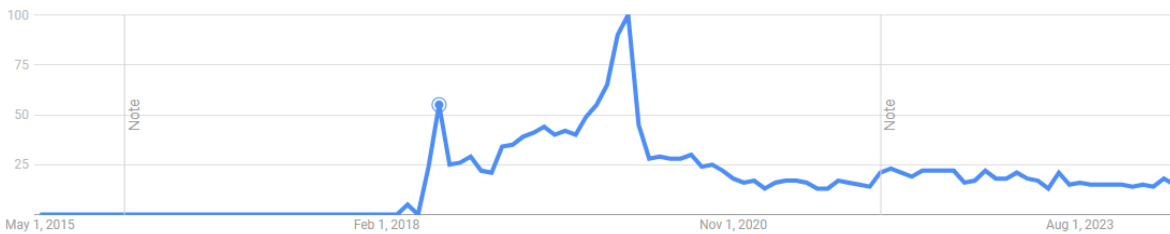


Figure OA2a. Google Search Trends about "California Consumer Privacy Act"



Figure OA2b. Google Search Trends about "Consumer Privacy"

Figure OA3. CFPB complaints

These figures display the density of complaints related to the improper use of consumer reports, both before and after the enactment of DPL. The data is from the Consumer Financial Protection Bureau (CFPB) and covers the years 2018-2023. The x-axis represents time, such as the number of months relative to the enactment of the privacy law, with a red dashed line marking the point of the law's enactment. The y-axis represents the density of complaints. Figure A2a illustrates the complaints received by lenders in the sample regarding the improper use of consumer reports before and after the enactment of privacy protection laws. Figure A2b compares complaints about the improper use of consumer reports between states that were about to enact privacy protection laws (marked as 0) and neighboring states that did not enact such laws (marked as 1).

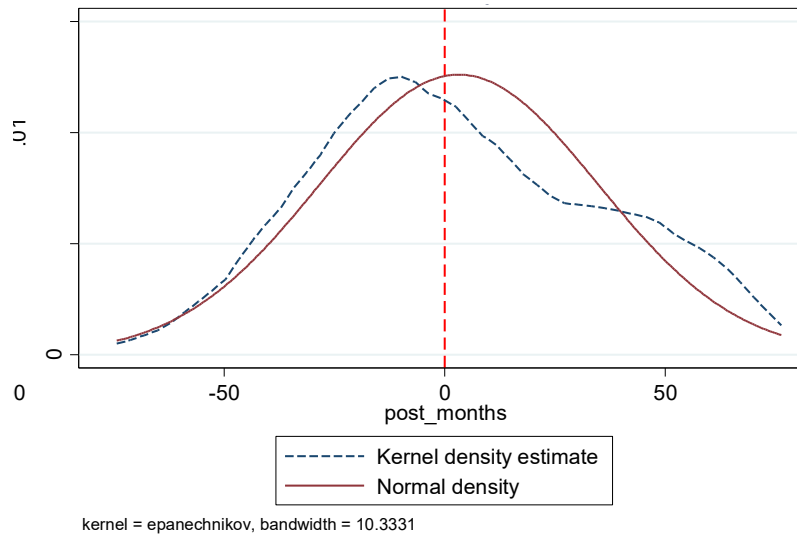


Figure OA3a. Distribution of "improper use of your report" complaints

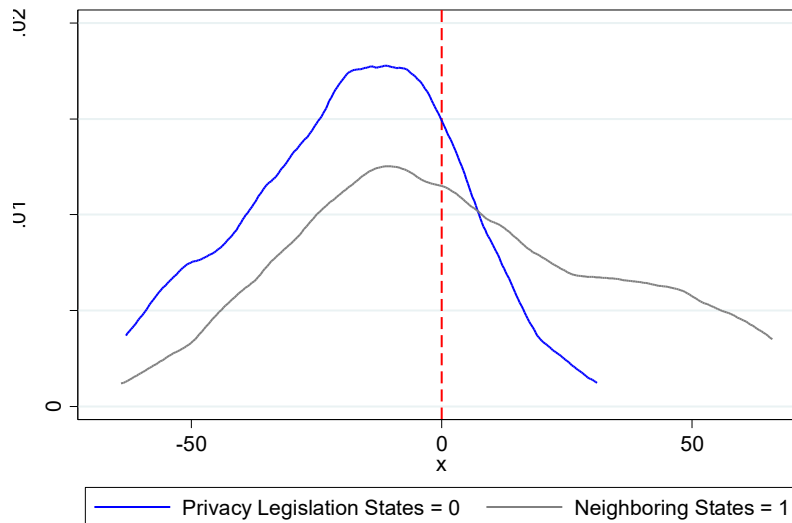


Figure OA3b. Distribution of "improper use of your report" complaints by DPL and non-DPL states

Figure OA4. DTI Distribution by Privacy Law Timing

This figure presents the distribution of Debt-to-Income (DTI) ratios for minority applicants and non-minority applicants, separated by the timing of DPL. The x-axis shows the DTI ratio, while the y-axis indicates the fraction of observations in each bin. The bars represent the relative frequency of DTI values before (Pre, in gray) and after (Post, in light blue) the enactment of privacy laws. The figure provides a visual comparison of DTI distributions across the two periods.

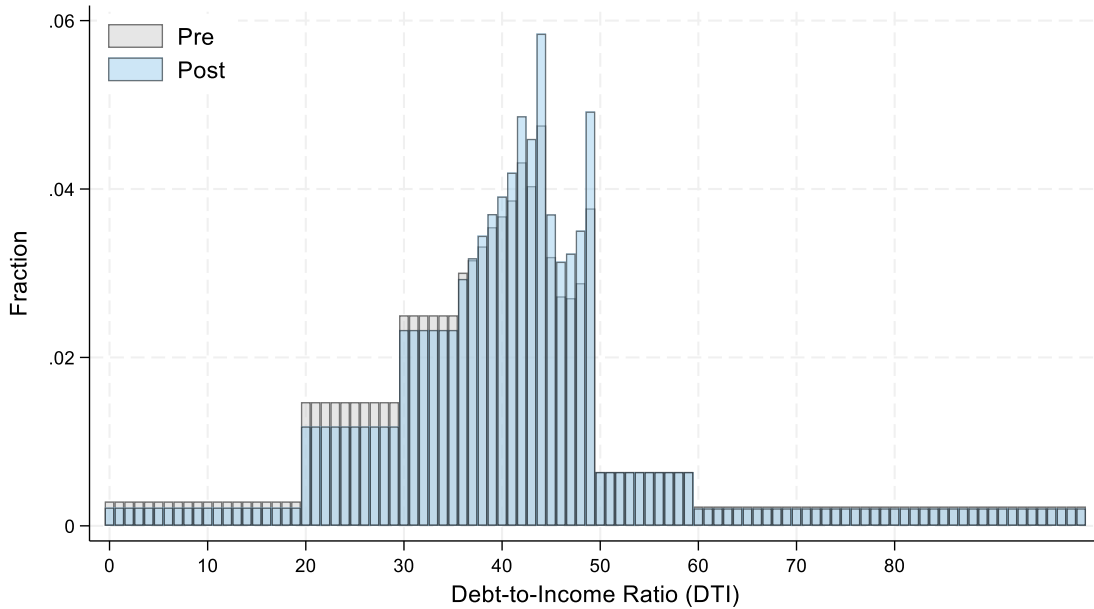


Figure OA4a. DTI Distribution by Privacy Law Timing of Minority

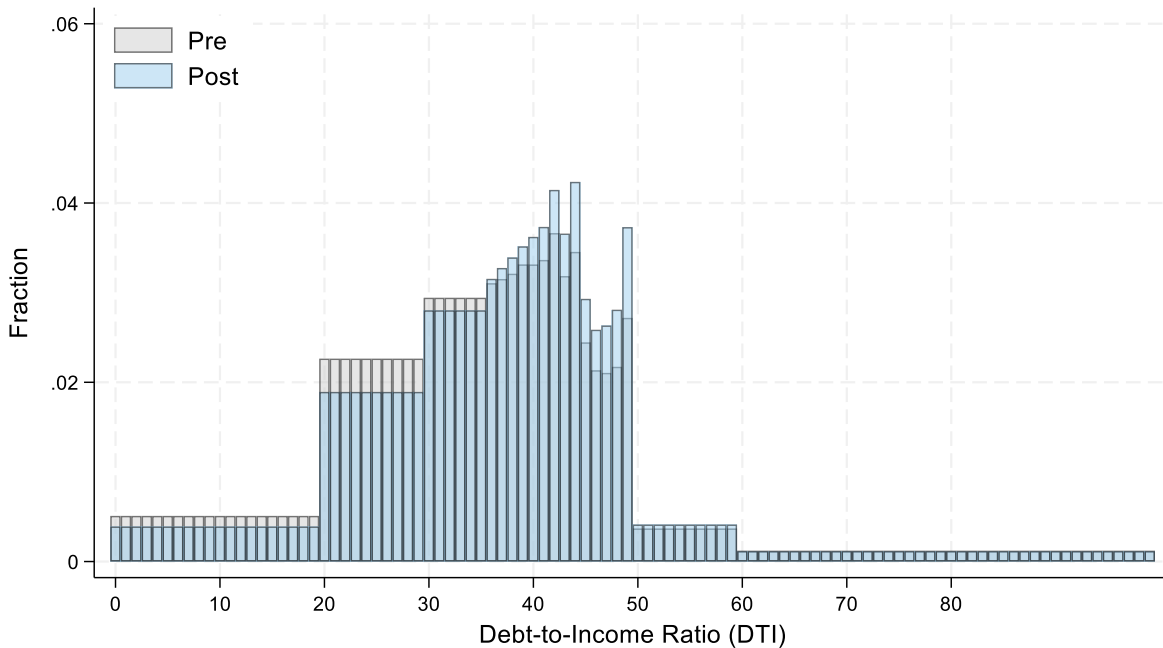


Figure OA4b. DTI Distribution by Privacy Law Timing of Non-Minority

Figure OA5. IAT data distribution

These figures display the county-level distribution of these bias measures, each standardized to have zero mean and unit variance. These data are derived from the Implicit Association Test (IAT) and utilize measures of both implicit and explicit biases, data derived from the Implicit Association Test (IAT) database (<https://osf.io/gwofk/>). Each test measures biases against specific racial groups (e.g., Black, White), defining implicit and explicit biases as preferential or prejudicial emotional responses toward White and Black individuals. The darker the color and the larger the labeled number, the stronger the preference for White individuals and the bias against Black people and other minorities. Lighter colors and smaller labeled numbers indicate a stronger preference for Black individuals and other minorities. Colors that are intermediate and numbers that are closer to zero suggest a near absence of bias, indicating no significant preference for either Black or White individuals.

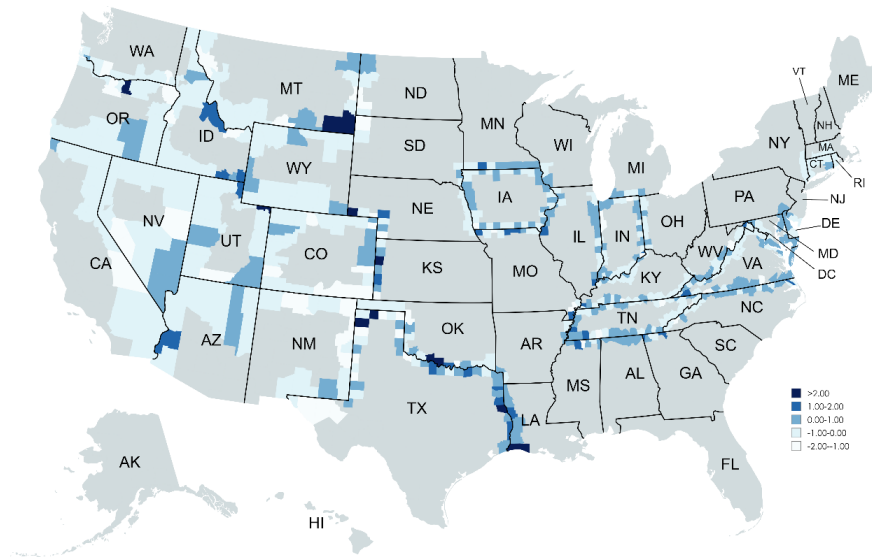


Figure A5a. County-level distribution of explicit racial animus

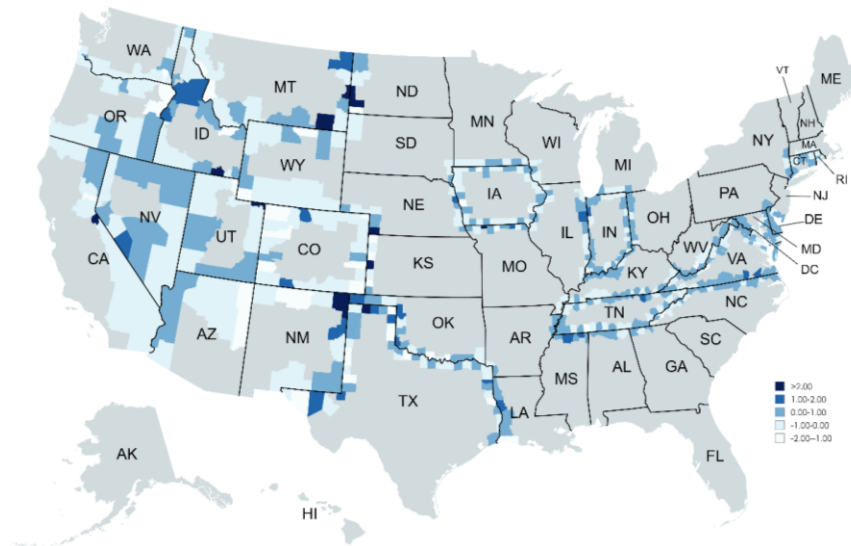


Figure A5b. County-level distribution of implicit racial animus

Figure OA6. Dynamic Effects of DPL on Mortgage Racial Gap by Loan Purpose

These figures show the dynamic effects of DPL on interest rates around the enactment date of the DPL by different mortgage purposes (home purchase or refinancing). The horizontal axis presents the year relative to the year of the enactment of DPL, split into 3 years prior to enactment (*Pre_3*, *Pre_2*, *Pre_1*), 3 years after enactment (*Post_1*, *Post_2*, *Post_3*), and the same year *Post_0*. The vertical axis presents the difference in interest rates of minority and non-minority borrowers, estimated using Model (2). The bar chart represents a 95% confidence interval.

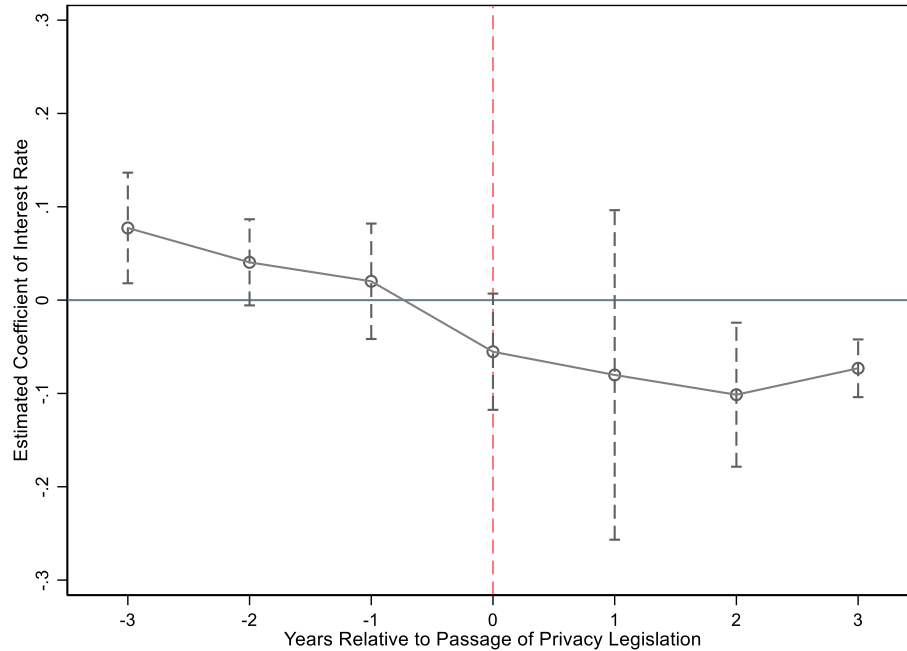


Figure OA6a. Effect on interest rate disparities (Home Purchase)

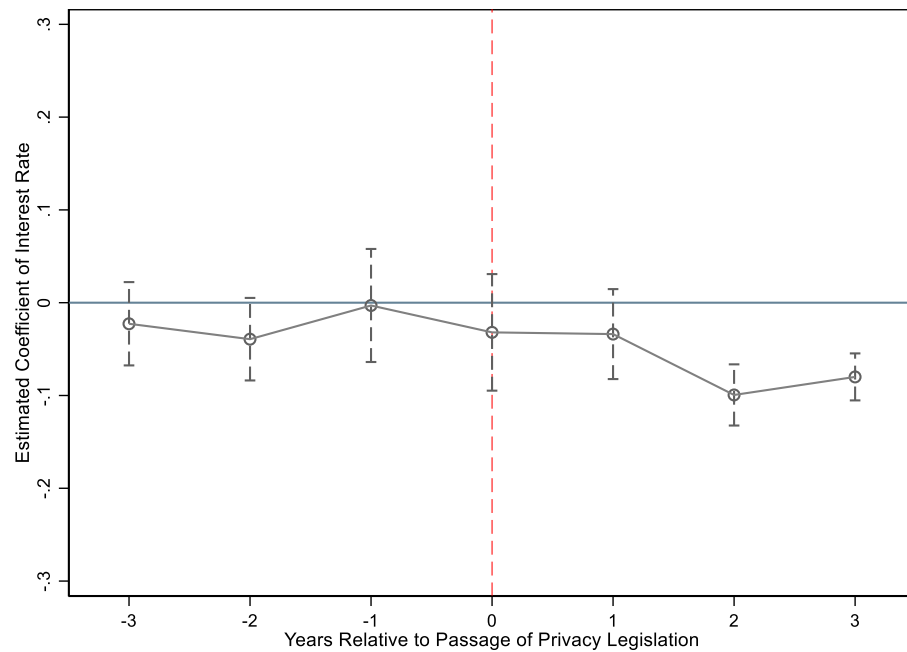


Figure OA6b. Effect on interest rate disparities (Refinancing)

Table OA1. Variable definitions

Variable	Definition
DPL	<i>DPL</i> indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed.
DPL_Effective	Equals 1 if the application <i>i</i> in year <i>t</i> at state <i>s</i> 's DPL is effective, 0 otherwise.
Minority	Equals 1 if the applicant's ethnicity is Latino, or their race is Black, 0 otherwise.
Female	Equals 1 if the applicant is female, 0 otherwise.
Denial	Whether lenders deny this mortgage application.
Interest Rate	The cost of mortgage loans as a percentage.
Interest Rate Spread	The difference between the loan's Annual Percentage Rate (APR) and the Average Prime Offer Rate (APOR) for a comparable transaction at the time the interest rate is set.
Delinquent	A dummy that equals one if an approved loan becomes 90-day delinquent or goes into foreclosure during the first 2 years of the loan's life
Credit Model	Whether a credit scoring model is used to generate the credit score or scores relied on in making the credit decision.
Debt_to_Income	The ratio, as a percentage, of the applicant's or borrower's total monthly debt to the total monthly income relied on in making the credit decision.
Loan_to_Value	The ratio of the total amount of debt secured by the property to the value of the property is relied on in making the credit decision.
Loan_to_Income	The ratio, as a percentage, of the applicant's or borrower's current amount of lending to the total monthly income relied on in making the credit decision.
Loan_Term	The amount of the covered loan, or the amount applied for.
Age	The age of the applicant is categorized into seven groups using numeric values from 1 to 7: age<25, 25-34, 35-44, 45-54, 55-64, 65-74, >74.
Gender Observed	Whether the gender of the applicant or borrower was collected based on visual observation or surname.
Race Observed	Whether the race of the applicant or borrower was collected based on visual observation or surname.
IAT bias	These data are derived from the Implicit Association Test (IAT) and utilize measures of both implicit and explicit biases, data derived from the Implicit Association Test (IAT) database (https://osf.io/gwofk/).
Fintech	Whether the lender is a Fintech or not indicates whether lending progress is online only or not (Buchak et al., 2018).
Bank	Whether the lender is a bank or not.

Table OA2. Alternative measure of lending cost

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on interest rate spread among minorities in the U.S. from 2018 to 2023. *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we use *Interest Rate Spread* to alternatively represent the cost of mortgage loans, which refers to the difference between the loan's Annual Percentage Rate (APR) and the Average Prime Offer Rate (APOR). For detailed information on the definition and construction of all control variables, please refer to Appendix Table OA1. Continuous variables, except macroeconomic variables, are winsorized at the first and 99th percentiles. Columns (1)–

(2) are the mortgage application samples from *counties* adjacent to states where DPL has been enacted versus counties adjacent to states where it has not been enacted. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Interest Rate Spread			
	(1)	(2)	(3)	(4)
DPL	0.017 (1.27)	0.017 (1.35)	0.014 (1.32)	0.012* (1.86)
Minority	0.136*** (14.57)	0.133*** (14.09)	0.115*** (8.36)	0.071*** (6.38)
DPL× Minority	-0.082*** (-5.18)	-0.082*** (-5.55)	-0.051*** (-3.72)	-0.034*** (-2.89)
Credit Model		-0.446*** (-8.37)		0.014 (0.62)
Debt_to_Income		0.047*** (16.20)		0.025*** (6.09)
LTV		0.051 (0.59)		0.378*** (24.07)
Loan_to_Income		-0.008*** (-6.37)		-0.027*** (-21.44)
Loan_Term		-0.001*** (-3.46)		-0.000*** (-10.53)
Race Observed		0.018 (1.20)		0.023** (2.55)
Age		-0.008 (-1.45)		0.019*** (14.89)
Credit Score				-0.002*** (-31.42)
Constant	0.386*** (113.08)	0.986*** (9.95)	0.274*** (118.55)	1.970*** (44.18)
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	5,638,515	5,632,768	150,922	150,922
R-squared	0.377	0.385	0.283	0.360

Table OA3. Classification into new and existing lenders

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on mortgage market racial disparities in the United States from 2018 to 2023, along with a subsample analysis distinguishing between existing lenders and new entrants. The Columns (1)–(2) display results for existing lenders (those that provided loans in each state before the enactment of privacy protection laws), while the Columns (3)–(4) present data for new entrants (those that began offering loans after these laws were enacted in each state). *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we primarily use dummy variable *Denial* to represent whether this application has been rejected or not, while we use *Interest Rate* to represent the cost of mortgage loans. The control variables include *Credit Model*, *Debt_to_Income*, *Loan_to_Value*, *Loan_to_Income*, *Loan_Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix OA1. Continuous variables, except macroeconomic variables, are winsorized at the first and 99th percentiles. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Existing Lenders		New Lender	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	0.005** (2.29)	0.013 (0.55)	0.024*** (5.08)	-0.191*** (-5.42)
Minority	0.070*** (15.38)	0.039*** (3.56)	0.134*** (13.15)	0.124*** (7.08)
DPL× Minority	-0.029*** (-3.08)	-0.065*** (-2.77)	-0.073** (-2.25)	-0.065** (-2.26)
Controls	Y	Y	Y	Y
Lender FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
County FE	Y	Y	Y	Y
Observations	7,175,553	5,983,729	11,485	9,884
R-squared	0.209	0.650	0.163	0.752

Table OA4. DPL Effects by Race Observability

This table reports subsample analyses of the effects of DPL on mortgage interest rates and denial rates, based on whether an applicant's race is likely to be observable. Treat indicates whether the mortgage application was filed in a state that had enacted a DPL, Post indicates whether the application occurred after the enactment of the DPL in that state, and Minority denotes whether the applicant belongs to an ethnic minority group. The dependent variables are Denial, which equals 1 if an application was rejected, and Interest Rate, measuring the cost of the mortgage loan. Columns (1) and (2) correspond to applications in which race is more likely to be observed, while Columns (3) and (4) correspond to applications in which race is less likely to be observable. Control variables include *Credit Model*, *Debt_to_Income*, *Loan_to_Value*, *Loan_to_Income*, *Ln_Loan_Amount*, *Race Observed*, and *Age*. Coefficients for control variables are omitted for brevity. Definitions of all variables and details of their construction are provided in Appendix Table OA1. T-statistics are reported in parentheses. Standard errors are clustered at the state level. Lender, year, and county fixed effects are included. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Race observed		Race unobserved	
	Denial	Interest Rate	Denial	Interest Rate
	(1)	(2)	(3)	(4)
DPL	0.007 (1.19)	-0.073 (-1.06)	0.003 (1.54)	0.021 (1.34)
Minority	0.112*** (16.65)	0.155*** (8.04)	0.081*** (13.98)	0.101*** (10.38)
DPL × Minority	-0.037** (-2.59)	-0.046 (-0.63)	-0.033*** (-3.45)	-0.081*** (-4.26)
Control Variables	Y	Y	Y	Y
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	261,316	205,513	6,842,963	5,736,291
R-squared	0.252	0.688	0.227	0.697

Table OA5. Differentiation by Race and Ethnicity

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on interest rate and denial among minorities, differentiating between Black and Hispanic borrowers by including dummy variables *Black* and *Hispanic*. *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed. For the dependent variable, we primarily use dummy variable *Denial* to represent whether this application has been rejected or not, while we use *Interest Rate* to represent the cost of mortgage loans. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix OA1. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Black and Non-Black</i>		
	Denial	Interest Rate
	(1)	(2)
DPL	-0.000 (-0.18)	0.011 (0.64)
Black	0.097*** (11.76)	0.123*** (11.80)
DPL × Black	-0.030*** (-2.72)	-0.059* (-1.71)
Control Variables	Y	Y
Year × Lender FE	Y	Y
County × Lender FE	Y	Y
Observations	6,857,157	5,759,868
R-squared	0.220	0.693
<i>Panel B. Hispanic and Non-Hispanic</i>		
	Denial	Interest Rate
	(1)	(2)
DPL	-0.001 (-0.72)	0.006 (0.29)
Hispanic	0.051*** (13.16)	0.076*** (7.45)
DPL × Hispanic	-0.023*** (-4.05)	-0.077*** (-6.54)
Control Variables	Y	Y
Year × Lender FE	Y	Y
County × Lender FE	Y	Y
Observations	6,631,909	5,569,517
R-squared	0.219	0.706

Table OA6. Robustness_ State-specific Effects of Privacy Laws

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on interest rate and denial among minorities, differentiating between individual states (Panel A) and states that include customer protection and/or data transparency. Panel A classifies states based on whether their DPL emphasizes Customer Protection (CP) or Transparency (TP) and provides the political leaning based on the previous three election results in 2024, 2020, and 2016 (D-Democratic, R-Republican). If a state's privacy law highlights "Customer Protection", the indicator variable equals 1; otherwise, it is 0. If a state's privacy law highlights "Transparency", the indicator variable equals 1; otherwise, it is 0. Panel B and Panel C show the subsample results of State-specific DPL. To be more specific, Panel B examines the effects of privacy laws emphasizing "Customer Protection" on interest rates and denial rates for minority and non-minority borrowers before and after enactment, comparing affected states with their neighboring states. Panel C analyses the effects of privacy laws focusing on "Transparency" on interest rates and denial rates for minority and non-minority borrowers before and after enactment, again comparing affected states with their neighboring states. For the dependent variable, we primarily use dummy variable *Denial* to represent whether this application has been rejected or not, while we use *Interest Rate* to represent the cost of mortgage loans. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix OA1. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Effects of privacy laws by state</i>								
No.	Privacy Date	State	Obs.	Denial	Interest Rate	Polit	CP	TP
(1)	Jun 28, 2018	California	1,258,010	-0.009*** (-9.64)	-0.038*** (-16.43)	D	1	1
(2)	Jul 7, 2021	Colorado	267,257	-0.008 (-1.23)	-0.069*** (-3.81)	D	1	1
(3)	May 10, 2022	Connecticut	966,940	0.012 (1.78)	-0.003 (-0.12)	D	1	1
(4)	Mar 24, 2022	Utah	237,971	0.029*** (8.29)	0.010 (0.37)	R	1	0
(5)	Mar 2, 2021	Virginia	1,093,878	-0.013** (-2.60)	-0.017 (-1.75)	D	1	1
(6)	Sep 11, 2023	Delaware	421,542	0.004 (0.50)	-0.107*** (-4.44)	D	1	0
(7)	May 1, 2023	Indiana	1,422,136	-0.006 (-1.61)	-0.119*** (-16.35)	R	0	1
(8)	Mar 28, 2023	Iowa	287,753	-0.024*** (-10.04)	-0.054** (-2.54)	R	1	0
(9)	May 19, 2023	Montana	88,753	-0.024*** (-8.15)	0.001 (0.09)	R	0	1
(10)	Jun 22, 2023	Oregon	489,581	-0.007** (-2.35)	-0.005 (-0.32)	D	1	1
(11)	May 11, 2023	Tennessee	562,770	-0.001 (-0.25)	-0.256*** (-11.83)	R	1	0
(12)	Jun 18, 2023	Texas	236,062	-0.023** (-3.20)	-0.073*** (-6.14)	R	0	1

(continued next page)

Table OA6. Robustness_ State-specific Effects of Privacy Laws (Cont'd)

<i>Panel B. Effects of privacy laws emphasizing "Customer Protection"</i>				
	Denial		Interest Rate	
	(1)	(2)	(3)	(4)
DPL (CP)	0.003 (1.45)	0.003 (1.53)	0.003 (1.45)	0.022 (1.15)
Minority	0.082*** (12.33)	0.081*** (11.75)	0.083*** (14.37)	0.099*** (8.82)
DPL (CP) × Minority	-0.034*** (-3.41)	-0.035*** (-3.65)	-0.035*** (-3.66)	-0.077*** (-3.65)
Borrower Controls	Y	Y	Y	Y
State Controls	N	Y	N	Y
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	6,233,002	6,000,363	7,111,837	5,019,196
R-squared	0.223	0.225	0.226	0.707
<i>Panel C. Effects of privacy laws emphasizing "Transparency"</i>				
	Denial		Interest Rate	
	(1)	(2)	(3)	(4)
DPL (TP)	0.004* (1.74)	0.005 (1.64)	0.005** (2.04)	0.022 (1.18)
Minority	0.082*** (12.61)	0.082*** (12.03)	0.083*** (14.46)	0.097*** (9.17)
DPL (TP) × Minority	-0.035*** (-3.65)	-0.036*** (-3.90)	-0.036*** (-3.87)	-0.076*** (-3.88)
Borrower Controls	Y	Y	Y	Y
State Controls	N	Y	N	Y
Year × Lender FE	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y
Observations	6,257,734	6,025,087	7,111,837	5,028,791
R-squared	0.226	0.228	0.226	0.708

Table OA7. Robustness_ Effects of Privacy Legislation on Racial Disparities (State-Level Sample)

This table presents the effects of DPL on mortgage denial rates and interest rates using the full state-level sample, rather than the adjacent-county design employed in earlier analyses. Specifically, we examine two samples: (i) states where DPL has been signed into law and their neighboring states (see Figure 1a), and (ii) states where DPL has become effective and their neighboring states (see Figure OA1). Panel A includes mortgage application data from states where DPL has been signed into law and from adjacent states. Panel B includes applications from states where the legislation is in effect and their neighboring states. The key variable, *DPL_Effective*, indicates the year *t* when the law became effective, in contrast to the signing date used in earlier analyses (e.g., Table 2). Control variables include *Credit Model*, *Debt_to_Income*, *Loan_to_Value*, *Loan_to_Income*, *Loan_Term*, *Race_Observed*, and *Age*. For brevity, coefficients on these controls are omitted from the table. Definitions and construction details for these variables are provided in Appendix OA1. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Effect of DPL on Racial Disparities full state sample</i>				
	Denial		Interest Rate	
	(1)	(2)	(3)	(4)
DPL	0.005** (2.41)	0.006** (2.58)	-0.045 (-0.91)	-0.040 (-0.85)
Minority	0.093*** (26.82)	0.072*** (23.00)	0.127*** (14.90)	0.057*** (7.05)
DPL × Minority	-0.029*** (-3.78)	-0.029*** (-3.50)	-0.088*** (-4.00)	-0.087*** (-4.14)
Control Variables	N	Y	N	Y
Lender FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
County FE	Y	Y	Y	Y
Observations	35,015,971	35,015,971	29,303,074	29,303,074
R-squared	0.178	0.212	0.656	0.669
<i>Panel B. Effect of effective DPL on Racial Disparities full state sample</i>				
	Denial		Interest Rate	
	(1)	(2)	(3)	(4)
DPL_Effective	0.001 (0.25)	0.001 (0.55)	0.051** (2.42)	0.047** (2.14)
Minority	0.101*** (18.41)	0.077*** (16.36)	0.119*** (12.05)	0.051*** (5.90)
DPL_Effective × Minority	-0.042*** (-7.28)	-0.040*** (-7.75)	-0.076*** (-4.92)	-0.077*** (-4.10)
Control Variables	N	Y	N	Y
Lender FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
County FE	Y	Y	Y	Y
Observations	19,372,134	19,372,134	15,867,976	15,867,976
R-squared	0.173	0.214	0.646	0.663

Table OA8. Robustness_ Lag effect of privacy legislation

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on interest rate, interest rate spread, and denial among minorities in the United States from 2018 to 2023. *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, we lagged the Post in these regressions. *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we primarily use dummy variable *Denial* to represent whether this application has been rejected or not, while we use *Interest Rate* to represent the cost of mortgage loans and *Interest Rate Spread* refers to the difference between the loan's Annual Percentage Rate (APR) and the Average Prime Offer Rate (APOR). The control variables include *Credit Model*, *Debt to Income*, *Loan to Value*, *Loan to Income*, *Loan Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix OA1. Continuous variables, except macroeconomic variables, are winsorized at the first and 99th percentiles. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Denial	Interest Rate	Interest Rate Spread
VARIABLES	(1)	(2)	(3)
DPL-1	0.004 (1.64)	0.033 (0.83)	0.036 (1.15)
Minority	0.083*** (14.21)	0.103*** (10.32)	0.133*** (14.01)
DPL-1 × Minority	-0.043*** (-5.26)	-0.088*** (-4.85)	-0.081*** (-5.89)
Control Variables	Y	Y	Y
Year × Lender FE	Y	Y	Y
County × Lender FE	Y	Y	Y
Observations	6,621,587	5,543,430	5,251,316
R-squared	0.224	0.680	0.387

Table OA9. Robustness_ Exclude Privacy legislation current year

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on interest rate, interest rate spread, and denial among minorities in the United States from 2018 to 2023 but excludes the current year of the DPL signed into law. *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we primarily use dummy variable *Denial* to represent whether this application has been rejected or not, while we use *Interest Rate* to represent the cost of mortgage loans and *Interest Rate Spread* refers to the difference between the loan's Annual Percentage Rate (APR) and the Average Prime Offer Rate (APOR). The control variables include *Credit Model*, *Debt to Income*, *Loan to Value*, *Loan to Income*, *Loan Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix OA1. Continuous variables, except macroeconomic variables, are winsorized at the first and 99th percentiles. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Denial	Interest Rate	Interest Rate Spread
	(1)	(2)	(3)
DPL	0.004 (1.62)	0.033 (0.83)	0.036 (1.15)
Minority	0.083*** (14.21)	0.103*** (10.32)	0.133*** (14.01)
DPL × Minority	-0.043*** (-5.26)	-0.088*** (-4.85)	-0.081*** (-5.89)
Control Variables	Y	Y	Y
Year × Lender FE	Y	Y	Y
County × Lender FE	Y	Y	Y
Observations	6,621,587	5,543,430	5,251,316
R-squared	0.224	0.680	0.387

Table OA10. Robustness_ Exclude Covid period 2020-2021

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on denial rate, interest rate, and interest rate spread among minorities in the United States from 2018 to 2023 but excludes the Covid period (2020 and 2021). *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we primarily use dummy variable *Denial* to represent whether this application has been rejected or not, while we use *Interest Rate* to represent the cost of mortgage loans and *Interest Rate Spread* refers to the difference between the loan's Annual Percentage Rate (APR) and the Average Prime Offer Rate (APOR). The control variables include *Credit Model*, *Debt to Income*, *Loan to Value*, *Loan to Income*, *Loan Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix OA1. Continuous variables, except macroeconomic variables, are winsorized at the first and 99th percentiles. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Denial	Interest Rate	Interest Rate Spread
	(1)	(2)	(3)
DPL	0.001 (0.19)	0.016 (0.57)	0.015 (0.71)
Minority	0.091*** (13.84)	0.109*** (7.95)	0.129*** (10.21)
DPL × Minority	-0.032*** (-3.07)	-0.096*** (-3.73)	-0.090*** (-4.98)
Control Variables	Y	Y	Y
Year × Lender FE	Y	Y	Y
County × Lender FE	Y	Y	Y
Observations	3,545,046	2,852,391	2,678,855
R-squared	0.243	0.536	0.370

Table OA11. Robustness_ Exclude the 10 largest lenders

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on interest rate, interest rate spread, and denial among minorities in the United States from 2018 to 2023. We exclude the top ten lenders (which have received the most applications). *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we primarily use dummy variable *Denial* to represent whether this application has been rejected or not, while we use *Interest Rate* to represent the cost of mortgage loans and *Interest Rate Spread* refers to the difference between the loan's Annual Percentage Rate (APR) and the Average Prime Offer Rate (APOR). The control variables include *Credit Model*, *Debt_to_Income*, *Loan_to_Value*, *Loan_to_Income*, *Loan_Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix OA1. Continuous variables, except macroeconomic variables, are winsorized at the first and 99th percentiles. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Denial	Interest Rate	Interest Rate Spread
	(1)	(2)	(3)
DPL	0.005** (2.61)	0.019 (1.20)	0.013 (1.14)
Minority	0.065*** (12.93)	0.082*** (7.42)	0.111*** (12.87)
DPL × Minority	-0.027*** (-2.81)	-0.070*** (-3.21)	-0.067*** (-4.26)
Control Variables	Y	Y	Y
Year × Lender FE	Y	Y	Y
County × Lender FE	Y	Y	Y
Observations	4,741,612	4,142,398	3,899,640
R-squared	0.202	0.709	0.380

Table OA12. Robustness_ PSM matching borrowers' credit quality and property

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on mortgage market racial disparities in the United States from 2018 to 2023, based on a propensity score matched sample. We applied propensity score matching (PSM) to identify the closest matches between loans in states with privacy protection legislation and those in states without such legislation. The matching criteria included information on collateral, types of mortgages and lien status, demographics, and all credit attributes listed in Table 2, i.e. loan-to-income ratio, loan-to-value ratio, debt-to-income ratio, applicant age, applicant race and mortgage value. *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we primarily use dummy variable *Denial* to represent whether this application has been rejected or not, while we use *Interest Rate* to represent the cost of mortgages. The control variables include *Credit Model*, *Debt_to_Income*, *Loan_to_Value*, *Loan_to_Income*, *Loan_Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix OA1. Continuous variables, except macroeconomic variables, are winsorized at the first and 99th percentiles. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)
	Denial	Interest Rate
DPL	0.007*** (4.38)	-0.008 (-0.29)
Minority	0.073*** (19.79)	0.051*** (4.55)
DPL× Minority	-0.032*** (-3.46)	-0.077*** (-3.42)
Controls	Y	Y
Lender FE	Y	Y
Year FE	Y	Y
County FE	Y	Y
Observations	4,671,597	3,888,245
R-squared	0.211	0.649

Table OA13. Robustness_ Different combinations of Fixed effects

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on denial rate and interest rate among minorities in the United States from 2018 to 2023. *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. For the dependent variable, we primarily use dummy variable *Denial* to represent whether this application has been rejected or not, while we use *Interest Rate* to represent the cost of mortgage loans. The control variables include *Credit Model*, *Debt_to_Income*, *Loan_to_Value*, *Loan_to_Income*, *Loan_Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix OA1. Continuous variables, except macroeconomic variables, are winsorized at the first and 99th percentiles. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

<i>Panel A. Different Combinations of Fixed Effects</i>				
	Denial		Interest Rate	
	(1)	(2)	(3)	(4)
DPL	0.010 (1.62)	0.009 (1.53)	0.530** (2.06)	0.024* (1.70)
Minority	0.076*** (17.15)	0.076*** (17.19)	0.135*** (6.78)	0.042*** (3.54)
DPL × Minority	-0.029*** (-3.32)	-0.029*** (-3.40)	-0.304*** (-2.78)	-0.067*** (-3.03)
Control Variables	Y	Y	Y	Y
Lender FE	Y	Y	Y	Y
Year FE	N	Y	N	Y
Observations	7,187,098	7,187,098	5,993,673	5,993,673
R-squared	0.207	0.208	0.203	0.649
<i>Panel B. Higher-Order Interactions of Fixed Effects</i>				
	Denial		Interest Rate	
	(5)	(6)	(7)	(8)
DPL	0.008 (1.28)		0.031** (2.10)	
Minority	0.075*** (17.29)	0.068*** (15.17)	0.042*** (3.84)	0.038*** (4.15)
DPL × Minority	-0.030*** (-3.45)	-0.030*** (-3.43)	-0.069*** (-3.17)	-0.051** (-2.62)
Control Variables	Y	Y	Y	Y
Lender × Year FE	Y	N	Y	N
Lender × Year × County FE	N	Y	N	Y
Observations	7,186,193	7,065,783	5,992,762	5,878,026
R-squared	0.222	0.257	0.677	0.700

Table OA14. Denial Reason Differentiations

This table presents the regression results and t-statistics (in parentheses) for the impact of DPL on the denial (reject) reason among minorities in the United States from 2018 to 2023, categorized by different denial reasons, i.e., Credit history, Insufficient cash (downpayment, closing costs), Debt-to-income ratio, Employment history, Collateral, Mortgage insurance denied. We only include samples that have been denied by lending institutions and provide specific reasons for denial. *DPL* indicates whether the mortgage application was submitted in a state where the data privacy legislation was signed, and *Minority* indicates whether the applicant belongs to an ethnic minority. The control variables include *Credit Model*, *Debt to Income*, *Loan to Value*, *Loan to Income*, *Loan Term*, *Race Observed*, and *Age*. The coefficients on all the control variables are omitted for brevity. For the definitions of all the control variables and the details of their construction, see Appendix Table OA1. Standard errors are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	Credit history	Insufficient cash (down payment, closing costs)	Debt-to- income ratio	Employment history	Collateral	Mortgage insurance denied
	(1)	(2)	(3)	(4)	(5)	(6)
DPL	0.009*** (2.83)	0.003 (1.27)	-0.003 (-0.97)	-0.001* (-1.96)	-0.003 (-1.19)	0.000 (0.21)
Minority	0.063*** (9.77)	0.004*** (2.91)	-0.000 (-0.03)	-0.001* (-1.99)	-0.016*** (-6.35)	0.001*** (3.91)
DPL × Minority	-0.019*** (-3.09)	-0.003* (-1.95)	0.006 (1.21)	-0.000 (-0.05)	-0.005 (-1.00)	-0.000 (-1.32)
Control Variables	Y	Y	Y	Y	Y	Y
Year × Lender FE	Y	Y	Y	Y	Y	Y
County × Lender FE	Y	Y	Y	Y	Y	Y
Observations	1,119,417	1,119,417	1,119,417	1,119,417	1,119,417	1,119,417
R-squared	0.244	0.136	0.265	0.081	0.155	0.098